Thermodynamics & Statistical Physics

2025/26/2

What is this course about?

Some central concepts: Temperature, Energy, Entropy
↑ 0th law ↑ 1st law ↑ 2nd law

Temperature: Most intuitive but how to quantify?
How does it affect other properties?

↳ most striking: Phase transitions
(e.g. ice - water - vapor

Energy: Inherited from classical mechanics

↳ only seems conserved in special cases (conservative forces)

Thermo: energy is always conserved

Conversion of energy between different forms → engines

"lost" energy → heat

Entropy: Quantifies reversibility / irreversibility

Very broad: applies to all (large) physical systems
from (bio) chemical reactions to engines
to stars and black holes

(Classical / Phenomenological) Thermodynamics:

- Focuses on macroscopic behavior, ignores micro. structure
- Basic postulates (laws 0, 1, 2, 3) derived from observations
  → explore consequences
- Material properties (heat capacity etc.) need to be measured

Statistical Physics

- Micro ⇒ Macro

  E.g.: A gas as a collection of molecules

- Deals with statistical (probabilistic) properties of large ensembles of particles
  - entropy plays a central role
    - rediscovered by Shannon in information theory

Both: Mostly systems in / close to equilibrium

↓

can be extended to non-equilibrium thermo. / stat. phys.

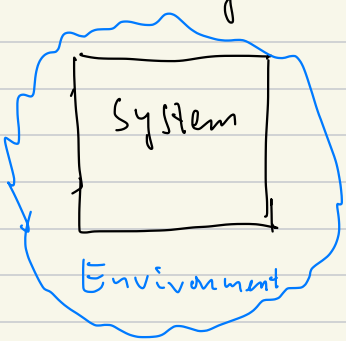
Thermal equilibrium, 0th law and temperature

Macroscopic physical system

e.g. $1\,m^3$ of water ($O(10^{23})$ atoms/molecules)

In general system is in contact with its environment/surroundings

Let's imagine that system is placed in some container



Different types of contact

depending on type of wall

a) Can wall move?  piston

b) Can particles go through? 

c) Can energy flow through?

A special type of wall: adiabatic → no contact with the environment

⇒ isolated system: 

Fact: An isolated system eventually (usually on a reasonable time scale) settles down to a (thermal) equilibrium state

↳ no more macroscopic changes

State variables

Equilibrium state described by only a few parameters

↳ volume, pressure, masses of different chemical components, ...
(V) (P) ($M_1, M_2, M_3, \ldots$)

↓

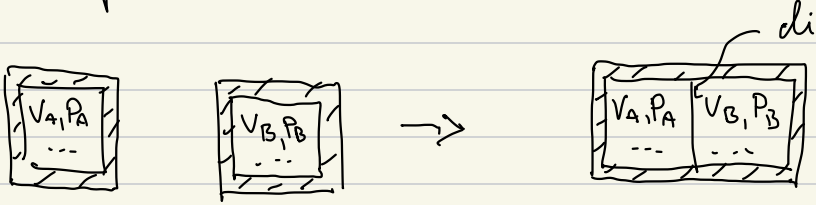
functions of these are called <u>state properties</u>

Diathermal wall: allows exchange of energy

↳ <u>thermal</u> contact

Consider 2 systems A and B

First allow both to reach equilibrium
then put them into thermal contact



Are A and B in thermal equilibrium with each other?

↳ only for particular choices of parameters

(e.g. keep $V_A, P_A, \ldots$ fixed and adjust $V_B, P_B, \ldots$)

notation: $A \sim B$

0th law: 3 systems A, B, C

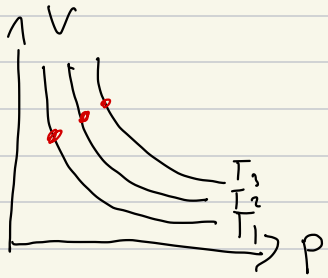
If C is in equilibrium with both A and B

$\Rightarrow$ A and B are also in equilibrium with each other:

$A \sim C$ and $B \sim C \Rightarrow A \sim B$

("Equilibrium is transitive")

Let us draw all systems that are in equilibrium with some fixed reference



0th law means that these lines cannot cross

$\Rightarrow$ We can define a state variable $T = f(P, V, \ldots)$

such that $T_A = T_B$ exactly when $A \sim B$

$\rightarrow$ **temperature**

T = const. curves: **isotherms**

$T = f(P, V, \ldots)$: **equation of state**

More formally:

equilibrium condition: $A \sim B \Leftrightarrow F(P_A, V_A, P_B, V_B) = 0$

now: $F_1(P_A, V_A, P_C, V_C) = 0 \rightarrow P_C = f_1(P_A, V_A, V_C)$

$F_2(P_B, V_B, P_C, V_C) = 0 \rightarrow P_C = f_2(P_B, V_B, V_C)$

$A \sim C$ and $B \sim C \Rightarrow f_1(P_A, V_A, V_C) = f_2(P_B, V_B, V_C)$

$\Downarrow$ 0th law $\qquad \updownarrow$?

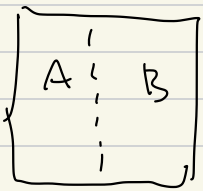
$F_3(P_A, V_A, V_B, V_B) = 0 \leftarrow$ does not contain V_C

$\Rightarrow$ must also cancel in $f_1 = f_2$

$\rightarrow f_1(P_A, V_A) = f_2(P_B, V_B) \quad (= f_3(P_C, V_C))$

$\Rightarrow$ empirical temperature $\boxed{T = f(P, V)}$ equation of state

Imagine dividing the system in two:



$V = V_A + V_B$
$M = M_A + M_B$
} extensive variables

$P = P_A = P_B$
$T = T_A = T_B$
} intensive variable

—

How to measure T?

↳ need to look at how it affects other quantities

Simple example: thermal expansion

↓

length / volume changes with T

→ e.g. mercury-in-glass thermometer

Modern thermometers: resistance / voltage

⇒ thermometric variable X

choose scale of T such that $X = c \cdot T_X + d$

Need reference points

i) 2-point scale → fixes c, d

e.g. Celsius: $T=0$: water freezing
$T=100$: -11- boiling
} at $P=P_{atm}$

ii) 1-point scale: set $d=0$

e.g. ideal gas scale ≡ Kelvin scale

In ideal gas, isotherms have $P \cdot V = \text{const.}$

(approx. true for real gases and improves as they get more dilute

↳ choose $P \cdot V = c \cdot T$

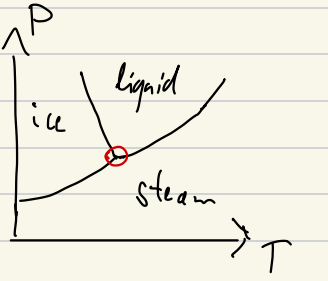
Still need 1 reference point

↳ choose triple point of water as $T = 273.16\,K$

(no need to specify pressure!)

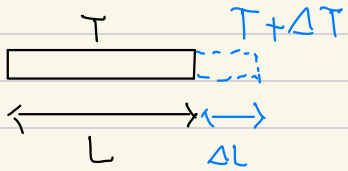
$c \sim N$ (no. of particles)

→ $PV = NRT$ ideal gas law



Thermal expansion:

1-dim. rod:



$$\Delta L \approx \alpha \cdot L \cdot \Delta T$$

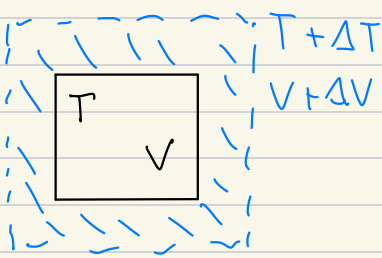
α: coefficient of (linear) thermal expansion

more generally: $\alpha := \frac{1}{L}\frac{dL}{dT}$

in general depends on T, but only weakly

$$L(T+\Delta T) = L(T)\,(1 + \alpha \Delta T + \alpha_2 \Delta T^2 + \dots)$$

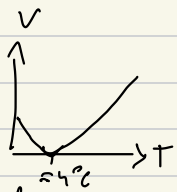
3-dim. volume:



$$\Delta V \approx \beta \cdot V \cdot \Delta T$$

$$\beta = \frac{1}{V}\frac{dV}{dT}$$ volume coefficient

Comments about thermal expansion:



- Usually $\beta_{liquid} > \beta_{solid}$

  Exception: water → surface of lakes freezes first

- Above assumes an isotropic body

  ↳ in crystal, α can be direction-dependent
  can even be < 0 in some directions

  ↳ can cause stress if crystal is heated in a non-uniform way

- Thermal expansion is why glass breaks from too hot liquid
  (actually, thick glass is worse!)
  (glasses used in optics need to be cooled very slowly)

- Can be some a problem in measuring length
  Low time in mechanical clocks based on pendula

  ↳ need to use material with small α (invar)
  (and 2 materials to compensate)

- Bimetallic strip: also makes thermometers, switches etc.

- Need to include gaps in bridges, train tracks, roads

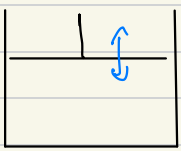
- Reinforced concrete works because $\beta_{steel} \approx \beta_{concrete}$

Thermodynamic processes

non-isolated system: changes in environment cause changes in state of system

→ thermodynamic process

example setup: gas in a container with movable piston



If we move the piston rapidly:

↳ gas is initially not in equilibrium (e.g. $P \neq$ const.)

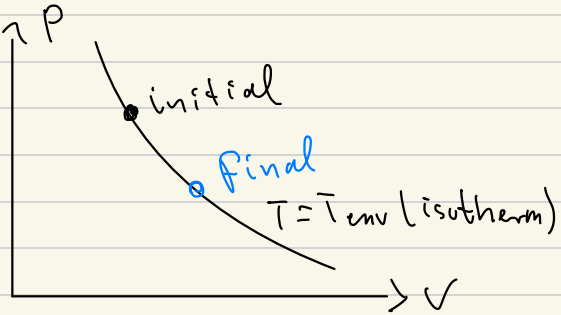
Eventually, if the piston stops moving and we wait long enough, it settles down to a new equilibrium state

↓

What this is, depends on how the process was performed

E.g. with diathermal walls: $T_{init} = T_{fin} = T_{env}$ (+ large environment)

but in-between initial and final state, the system is not in equil. and cannot be represented on P-V diagram!

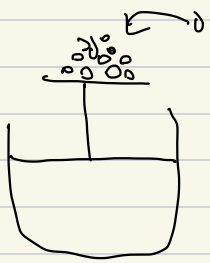


But it is often a very useful approximation to treat the system as going through a series of equilibrium states → described by curve on P-V diag.

Requires: $\tau_{change} \gg \tau_{relax}$

↳ <u>quasi-static process</u>

Quasi-static compression: $P \to P + \underbrace{dP}_{\text{infinitesimal}}$



"add 1 grain of sand at a time"

What happens if we reverse? $P \to P - dP$

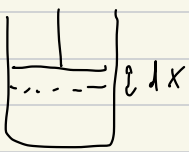
<u>reversible processes</u>: system returns to previous state with no change in the system <u>or</u> environment

→ requires that piston has zero friction

When we compress the gas: need to apply work

When gas expands: does work on weight

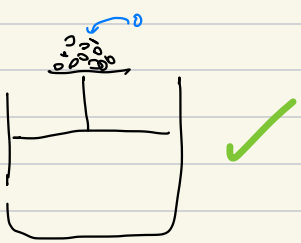
When process is reversible: $dW = -P \cdot A \cdot dx$, with $A \cdot dx = dV$



The $-$ sign is convention: measures work done *on* the system

$$\Rightarrow W_{1\to 2} = -\int_{V_1}^{V_2} P\,dV$$

(1) Also works for *some* non-reversible processes:



✓

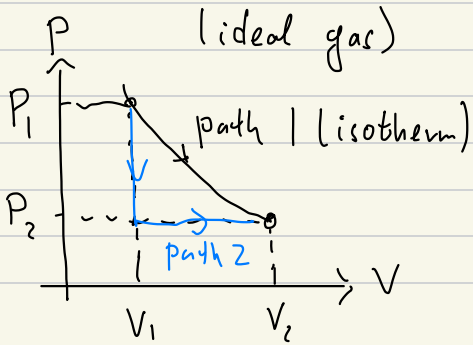


✗

(2) Depends on path:

$$W_{path\,1} = -\int_{V_1}^{V_2} dV\,\frac{nRT}{V} = -nRT\ln\left(\frac{V_2}{V_1}\right)$$

$$W_{path\,2} = -P_2(V_2 - V_1)$$



compare: $\phi = f(P,V)$ state function

$$d\phi = \frac{\partial f}{\partial P} dP + \frac{\partial f}{\partial V} dV$$

$$\int_A^B d\phi = f(P_B, V_B) - f(P_A, V_A)$$ independent of path!

Converse: assume we don't know $\phi = f(P,V)$,

we only know how to compute $\int d\phi$

pick arbitrary reference point (P_0, V_0)

define $\phi = f(P,V) = \int_{(P_0,V_0)}^{(P,V)} d\phi$

Two cases.

1) $\int d\phi$ indep. of path $\Leftrightarrow$ There exists $\phi = f(P,V)$

$d\phi$ is an <u>exact</u> differential

2) $\int dW$ depends on path $\Leftrightarrow$ no state function W

dW is <u>inexact</u> differential → notation: $đW$

We don't necessarily have endpoints parametrized by P, V

e.g. $(P_1, T_1) \to (P_2, T_2)$

↳ need to write $V = V(P, T)$

$$dV = \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT$$

$= -V \cdot \kappa$ $\quad$ $V \cdot \beta$

$$\kappa := -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_\kappa$$

(isothermal) compressibility

$$W = \int_{\text{path}} dW = -\int_{\text{path}} P\,dV = \int_{P_1}^{P_2} \kappa V P\,dP - \int_{T_1}^{T_2} \beta V P\,dT$$

In general everything (V, β, κ) depend on P, T

but dependence of β, κ is often weak

isothermal case: $T_1 = T_2 \to$ 2nd term vanishes

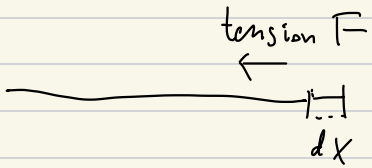
$$\Rightarrow W = \kappa \int_{P_1}^{P_2} P\,V(P)\,dP = \frac{\kappa}{2} V(P_*)\,(P_2^2 - P_1^2)$$

2nd mean value thm: $\int_{x_1}^{x_2} f(x)\,g(x)\,dx = g(x_*) \int_{x_1}^{x_2} f(x)\,dx$

$x_* \in [x_1, x_2]$

Other types of work:

a) Extensible wire



$$\delta W = F\,dx$$

b) surface film (e.g. soap bubble)



Surface tension : Γ

circumference : ℓ

$$\delta W = \Gamma \ell\, dx = \Gamma\, dA$$

c) Charging a battery: $\delta W = \mathcal{E}\, dz$ ($\mathcal{E}$: e.m.f. (electromotive force), z: electric field)

d) Dielectric material: $\delta W = \vec{E}\, d\vec{p}_e$ ($\vec{E}$: electric field, $\vec{p}_e$: electric dipole moment)

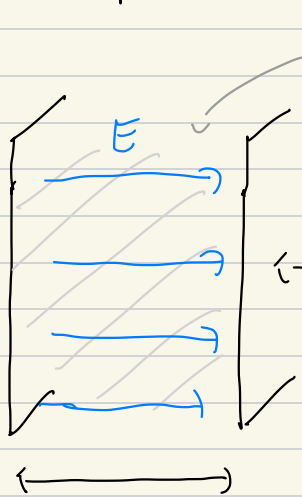
e) Magnetizable material: $\delta W = \vec{B}\, d\vec{M}$ ($\vec{B}$: magnetic field, $\vec{M}$: magnetic moment)
(e.g. para/diamagnet)

Comments: 1) All assumes reversible process

2) Always come in intensive-extensive pairs

E.g. Polarizable materials

capacitor:



charge with surface density $\sigma = \frac{Q}{A}$

Gauss: $D = \sigma$

$D = \varepsilon_0 E + P_{el}$

electric polarization density

voltage difference: $E \cdot \ell$

Increase $Q \to Q + dQ$

$$dW = E \ell \, dQ = E \underbrace{\ell A}_{V} d\sigma = E \cdot V \, dD$$

$$= E V (\varepsilon_0 dE + dP_{el})$$

$E V \varepsilon_0 dE$: work needed to charge the capacitor

$P_{el} = V P_{el}$ total dipole moment

$\Rightarrow dW = E \, dP_{el}$

Adiabatic process: system enclosed in adiabatic walls



$$\int dW_{adiab.} = U_{fin} - U_{in} = \Delta U$$

becomes independent of path!

Also indep. of type of work

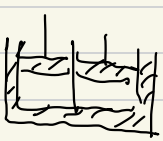
$$U_{tot} = U_{kin} + U_{pot} + U_{int} \qquad U_{int} = U_{int}(P,V,\ldots)$$

$$U(P,V,\ldots) = U(P_0,V_0,\ldots) + \int_{(P_0,V_0,\ldots)}^{(P,V,\ldots)} dW_{adiab}$$

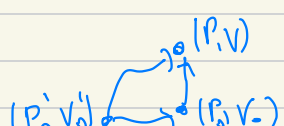
reference state

↑ defines U in thermo

Comments: ·) U is extensive



·) Dependence on ref. point:



$$U'(P,V) = \int_{(P_0',V_0')}^{(P,V)} dW_{ad.} = \int_{(P_0',V_0')}^{(P_0,V_0)} dW + \int_{(P_0,V_0)}^{(P,V)} dW = U(P,V) + \text{const.}$$

·) It must be possible to go between the two states adiabatically

↳ always true in at least one direction

If we remove adiabatic walls: energy can leak out!

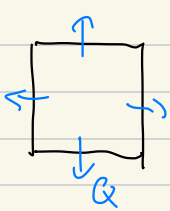
$\int đW \neq \Delta U$: this is why it becomes path-dependent

need to keep track of energy flowing out: heat

$\Delta U = W + Q$ (1st law)

$dU = đW + đQ$

dU: exact; $đW$, $đQ$: inexact



wall is characterized by its heat conductance

Adiabatic: conductance $= 0$

More generally: the smaller the conductance, the more time it takes to exchange heat with environment

↓

process is adiabatic if much faster than this time scale

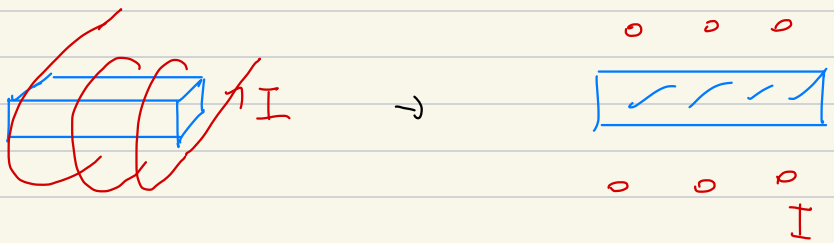
adiabatic quasi-static: slow compared to τ_{relax}

fast — " — τ_Q

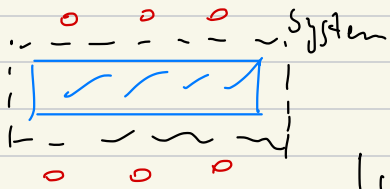
Forms of heat transport: conduction, convection, radiation

Comment: distinction between W and Q depends on where we draw the system's boundary

E.g. material surrounded by coils:



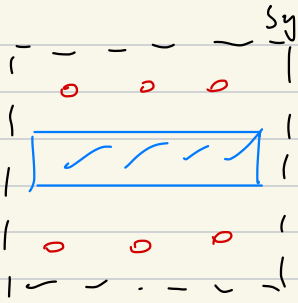
(1):



$\Delta U = Q$

(coils heat air etc.)

(2):



$\Delta U = W$

(work done by generator)

$V = \text{const.}$

when $W = 0$: $\Delta U = Q$ "isochoric process"

$dU = \delta Q_{\text{isochoric}}$

Expansion with constant pressure ("isobaric")

$$U_2 - U_1 = Q + W = Q - P(V_2 - V_1)$$

$$\rightarrow Q = (U_2 + PV_2) - (U_1 - PV_1)$$

def: $H := U + PV$ enthalpy

$$\rightarrow Q = H_2 - H_1 = \Delta H \quad ; \quad dH = \delta Q_{\text{isobaric}}$$

more generally ($P \neq \text{const.}$):

$$dH = d(U + PV) = dU + P\,dV + V\,dP = \delta Q + V\,dP$$

$dU = \delta Q - P\,dV$

Usually: larger U means higher T

→ heat capacity: $C := \lim_{\Delta T \to 0} \frac{Q}{\Delta T} = \frac{dQ}{dT}$

C is proportional to the amount of material

→ specific heat: $c = C/M$ characterizes the
or $c = C/n$ type of material

(n ↑ # moles)

also depends on process

·) isochoric specific heat: c_V ($V = \text{const.}$)

↳ $dW = -P\,dV = 0 \rightarrow dQ = dU \rightarrow c_V = \left(\frac{\partial U}{\partial T}\right)_V$

·) isobaric — " — : c_P ($P = \text{const.}$)

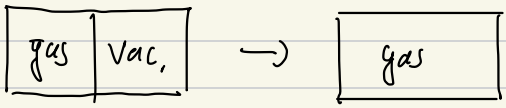
$dQ = dU + P\,dV \rightarrow c_P = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P$

$= \left(\frac{\partial H}{\partial T}\right)_P$ ($V \cdot \beta$)

Measuring c: method of mixtures

c_V and c_p in an ideal gas: $PV = nRT$ (n = # of moles)

Joule experiment:



no work $\rightarrow \Delta U = Q$

observation: no heating measured outside ($Q = 0$)

$\rightarrow U$ is independent of V at constant T

$U = U(T, V) = U(T)$

Comments: ·) We will show this later using 2nd law

·) Microscopically: ideal gas $\equiv$ non-interacting

↳ only kinetic energy, independent of V

$$dU = \frac{dU}{dT} dT = c_V dT \overset{!}{=} dQ - P dV$$

$$c_p = \frac{dQ_p}{dT} = \frac{c_V dT + P dV}{dT} = c_V + \left(\frac{\partial V}{\partial T}\right)_p P$$

$$= c_V + nR$$

work! > 0

Adiabatic process in an ideal gas

$$dQ = 0$$

$$\rightarrow 0 = \frac{dQ}{T} = \frac{c_V dT}{T} + \frac{P dV}{T} = c_V \frac{dT}{T} + nR \frac{dV}{V}$$

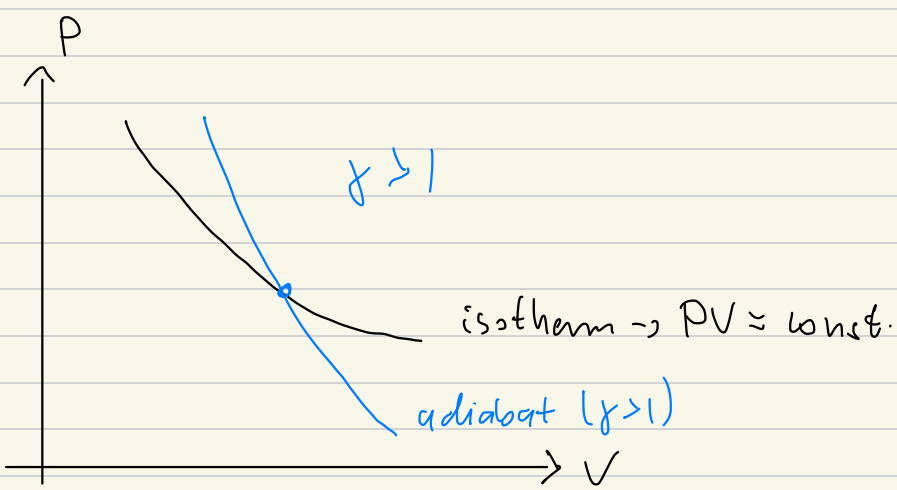
($c_p - c_V$ = nR; $\frac{dT}{T} = d(\ln T)$; $\frac{dV}{V} = d(\ln V)$)

$$\rightarrow c_V \ln T + (c_p - c_V) \ln V \simeq \text{const.}$$

$$\ln T + (\gamma - 1) \ln V = \text{const.} \qquad \gamma = \frac{c_p}{c_V}$$

$$= \ln (T V^{\gamma - 1}) \qquad \frac{T}{V} \sim P$$

$$\Rightarrow P V^{\gamma} = \text{const.}$$

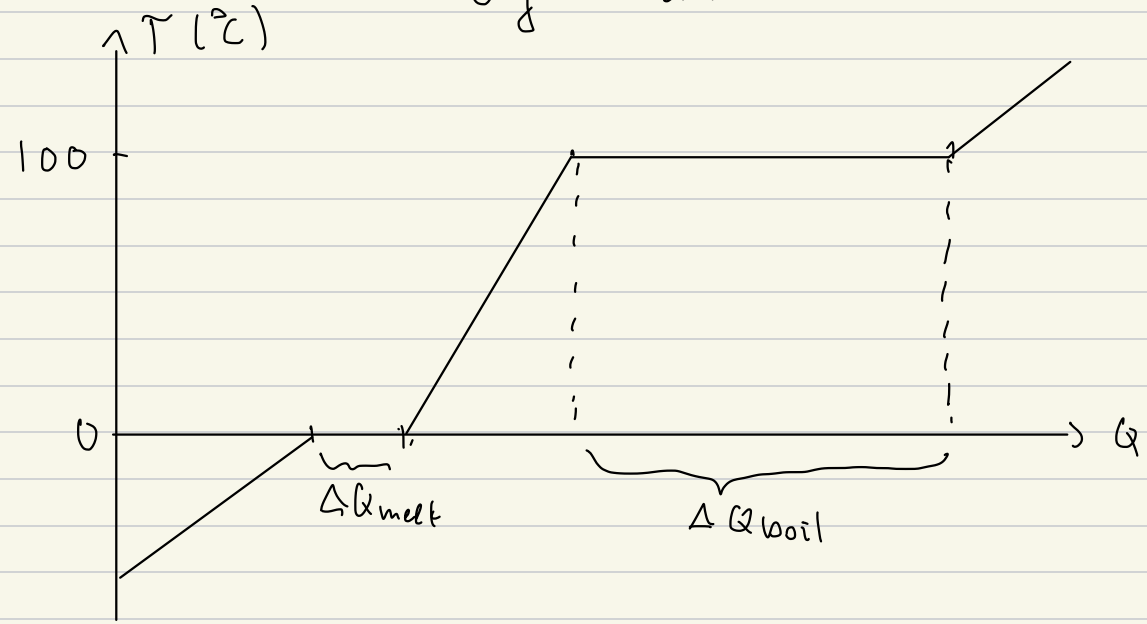


Latent heat:

Does Q always lead to increase in T?

No! → phase transitions

e.g. water:



ΔQ latent heat: proportional to mass M

specific latent heat: $L = \frac{\Delta Q}{M}$

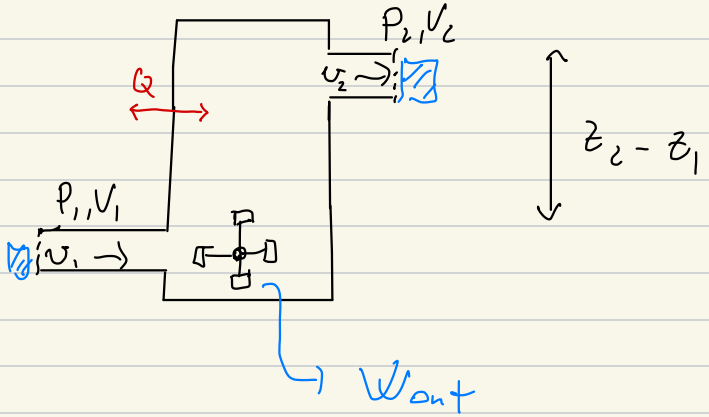
e.g. $L_{melt} \approx 333 \frac{kJ}{kg}$; $L_{boil} \approx 2260 \frac{kJ}{kg}$

(for water at atmospheric pressure)

In reverse process, heat flow to the environment

Application of 1st law in Flow processes

Setup:



$u_1 = u(P_1, V_1)$

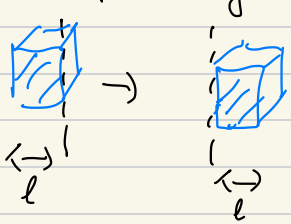
$u_2 = u(P_2, V_2)$

consider a piece of fluid with $M = 1$

$$W + Q = \Delta U_{tot} = \Delta U_{int} + \Delta U_{kin} + \Delta U_{pot}$$

$$\Delta U_{int} = u_2 - u_1 \qquad \Delta U_{kin} = \frac{1}{2}(v_2^2 - v_1^2) \qquad \Delta U_{pot} = g(z_2 - z_1)$$

consider pushing the fluid in at constant P_1



$$W = P_1 A \cdot l = P_1 V$$

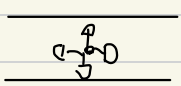
$$W = P_1 V_1 - P_2 V_2 - W_{out} \qquad H = U + PV$$

$$\Rightarrow W_{out} = H_1 - H_2 + \frac{1}{2}(v_1^2 - v_2^2) + g(z_1 - z_2) + Q$$

Special cases:

1) If flow is fast: $Q \approx 0$

2) Let $z_1 = z_2$:



$$W_{out} = H_1 - H_2 + \frac{1}{2}(v_1^2 - v_2^2)$$

3) Let also $W_{out} = 0$

e.g.: nozzle:



narrower tube → faster flow

What is v_2? → $v_2^2 = v_1^2 - 2(H_2 - H_1)$

Usually we know v_1, P_1, T_2, P_2

↳ need T_2 to determine H_2

Ideal gas approximation:

$$\left.\begin{aligned} TV^{\gamma-1} &= \text{const.} \\ V &= nR\frac{T}{P} \end{aligned}\right\} \rightarrow \frac{T^\gamma}{P^{\gamma-1}} = \text{const.} \rightarrow \frac{T_2^\gamma}{P_2^{\gamma-1}} = \frac{T_1^\gamma}{P_1^{\gamma-1}}$$

$$\rightarrow T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{1-1/\gamma}$$

Carnot cycle

Cyclic process: Initial state = final state

$\Delta U = 0 \rightarrow W = -Q$ ($W < 0$, $Q > 0$)

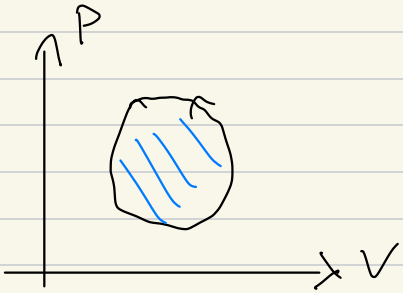
Do work by supplying heat

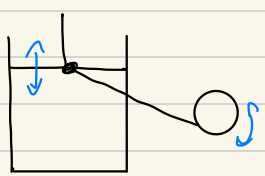
Cyclic → can be repeated

} (heat) engine!

If work comes from expansion

↳ W = area enclosed on P-V diagram





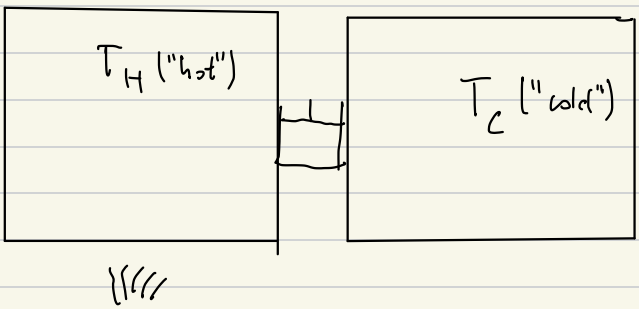
$Q = Q_{in} - Q_{out}$ → <u>efficiency</u>: $\eta := \frac{|W|}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}} \leq 1$

We can also have a cycle running backwards:

$W = Q_{out} - Q_{in} > 0$

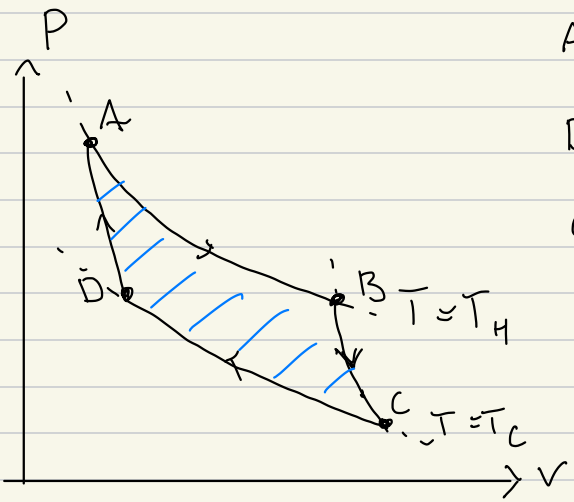
→ refrigerator: $\eta^R := \frac{Q_{in}}{W} = \frac{Q_{in}}{Q_{out} - Q_{in}}$ (coeff. of performance)

Carnot: Let the system only exchange heat in an isothermal way ($\Delta T = const.$)



$T_2 > T_1$

to maintain T_H



$A \to B$: $\Delta V > 0 \to W_{AB} < 0$; $Q_{AB} > 0$

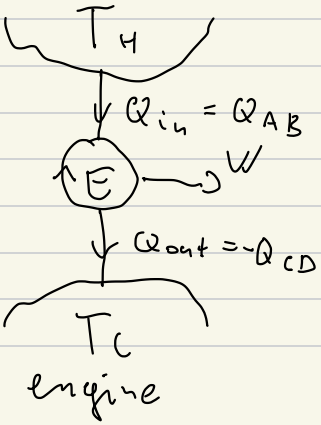
$B \to C$: $Q_{BC} = 0$ (adiabat)

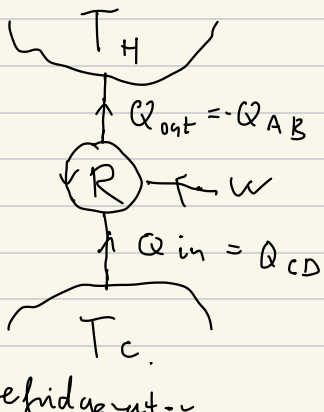
$C \to D$: $\Delta V < 0 \to W_{CD} > 0$; $Q_{CD} < 0$

$D \to A$: $Q_{DA} = 0$

$W_{ABCD} < 0$ (does work)

We can also run it in reverse: $W_{ADCB} > 0$





Assuming ideal gas:

$$U = U(T) \rightarrow U_A = U_B \rightarrow Q_{AB} = -W_{AB} = NRT_H \ln \frac{V_B}{V_A}$$

$$Q_{CD} = -W_{CD} = NRT_C \ln \frac{V_D}{V_C}$$

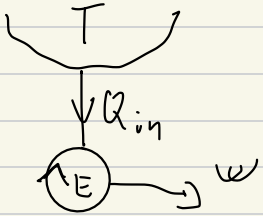
BC, DA are adiabats: $TV^{\gamma-1} = \text{const}$

$$\left.\begin{aligned} T_H V_B^{\gamma-1} &= T_C V_C^{\gamma-1} \\ T_H V_A^{\gamma-1} &= T_C V_D^{\gamma-1} \end{aligned}\right\} \left(\frac{V_D}{V_C}\right)^{\gamma-1} = \left(\frac{V_A}{V_B}\right)^{\gamma-1}$$

$$\eta = 1 - \frac{Q_{out}}{Q_{in}} = 1 - \frac{-Q_{CD}}{Q_{AB}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H} < 1$$

Note: $PV = NRT \rightarrow T \geq 0$!

Could we have $\eta = 1$?



$Q_{out} = 0 \rightarrow W = Q_{in}$

convert 100% of Q to W

(K): 2nd law (Kelvin): No ✗

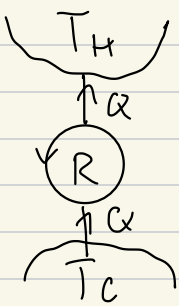
No perpetuum mobile (of 2nd kind)

(e.g. powering a car just be sucking heat out of the air)

Comments: •) cycle is important

•) Converting $W \rightarrow Q$ is fine

Similar question for refrigerators: $\eta^R = \infty$ $(W = 0)$



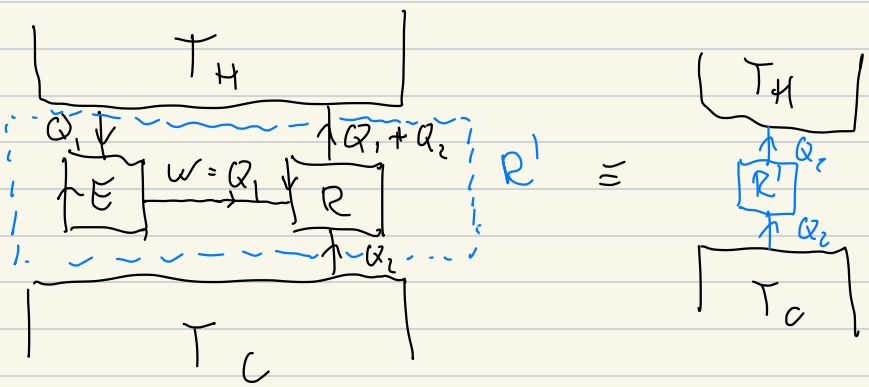
Heat flowing spontaneously from H to C

(C): Not possible ✗ (Clausius)

Equivalence of the two:

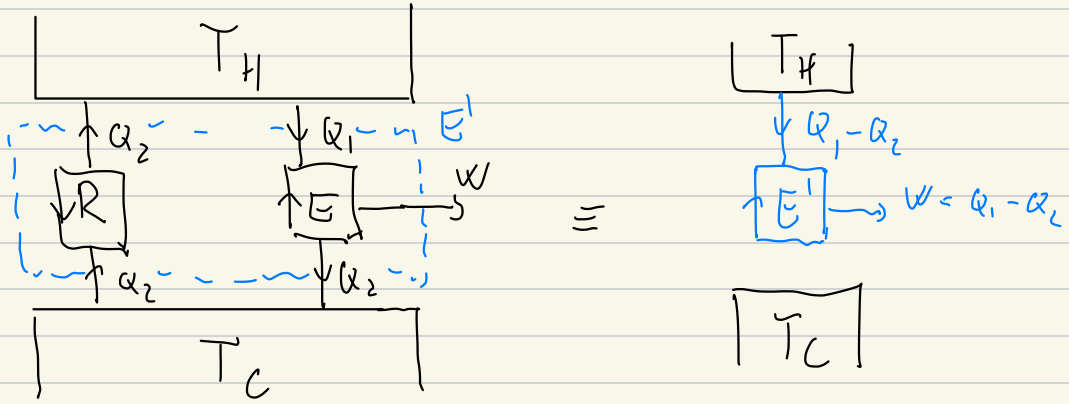
1) K is false $\Rightarrow$ C is false

use perfect E to create perfect R

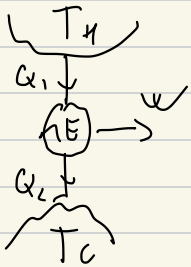


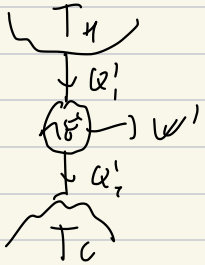
2) C is false $\Rightarrow$ K is false

use perfect R to create perfect E



Carnot's theorem: Let E be a reversible engine
E' arbitrary engine
operating between same T_H, T_C



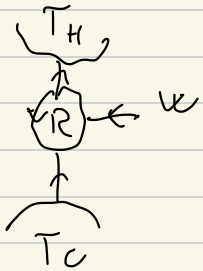


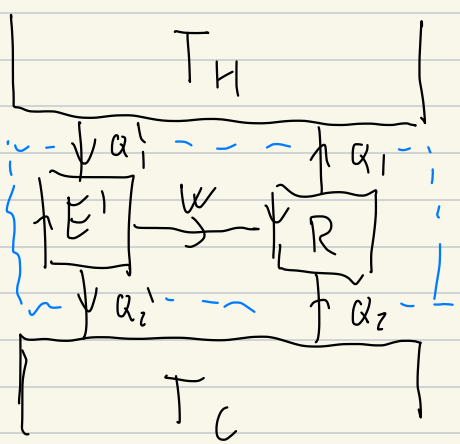
$\Rightarrow \eta_E \geq \eta_{E'}$

Proof: reverse E to get refrigerator R:

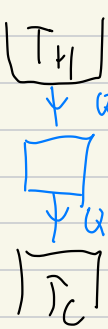
↓

combine with E' s.t. $W = W'$





$\equiv$



Assume $\eta_{E'} = \frac{W}{Q_1'} > \eta_E = \frac{W}{Q_1} \Rightarrow Q_1' < Q_1 \Rightarrow Q_1' - Q_1 < 0$

$\Rightarrow$ violates (C)

Corollary: If E, E' are both reversible: $\eta_E = \eta_{E'}$

All rev. engines have same η_{rev}

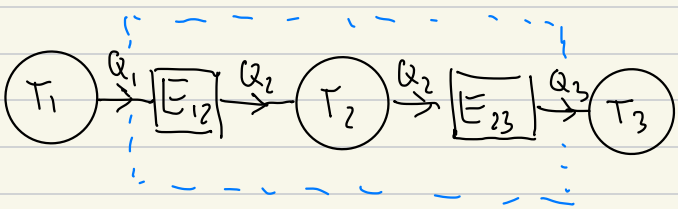
↳ depends only on T_H, T_C

$$\eta_{rev} = 1 - \frac{Q_{out}^{rev}}{Q_{in}^{rev}} = 1 - F(T_H, T_C)$$

but: $\frac{Q_2}{Q_1}\frac{Q_3}{Q_2} = \frac{Q_3}{Q_1} \rightarrow F(T_1, T_2)F(T_2, T_3) = F(T_1, T_3)$

T_2 must drop out!

2 engines "in parallel":

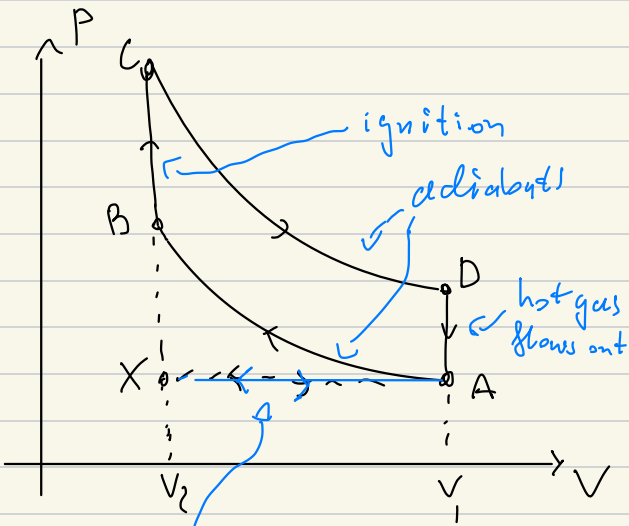


$$\Rightarrow F(T_1, T_2) = \frac{F(T_1)}{F(T_2)}$$ we can choose $F(T) = a \cdot T$

$$\rightarrow \eta_{rev} = 1 - \frac{T_H}{T_C}$$ defines thermodynamic temp. scale

but Carnot with ideal gas shows that thermodynamic scale ≡ ideal gas scale

Otto cycle: Toy model of internal combustion engine



$V = \text{const.} \rightarrow Q = C_V \Delta T$

$Q_{in} = C_{V_2} (T_C - T_B)$

$Q_{out} = C_{V_1} (T_D - T_A)$

We'll assume $C_{V_1} = C_{V_2}$

repel exhaust / draw in new air

adiabats: $TV^{\gamma-1} = \text{const.}$

$\rightarrow (T_D - T_A) V_1^{\gamma-1} = (T_C - T_B) V_2^{\gamma-1}$

$$\Rightarrow \eta = 1 - \frac{Q_{out}}{Q_{in}} = 1 - \frac{T_D - T_A}{T_C - T_B} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

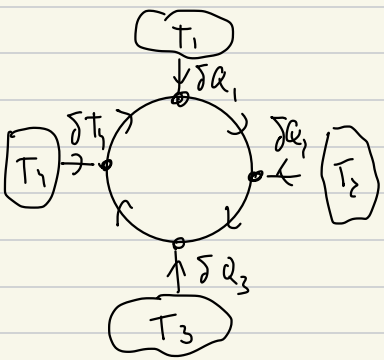
compression ratio: $r_c := V_2 / V_1$

typical car: $r_c = 7-8$

Formula-1 car: $r_c \approx 17$

Clausius inequality

cyclic process of system exchanging heat $Q_1, Q_2, \ldots$ with environments at temperatures $T_1, T_2, \ldots$



2nd law $\Rightarrow$ we cannot have all $\delta Q_i > 0$

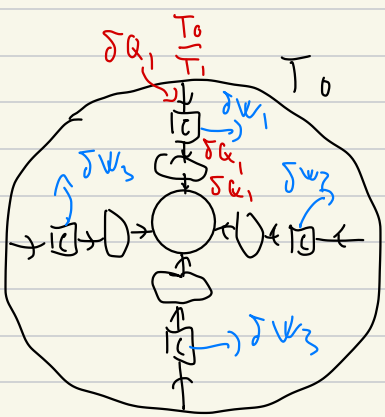
Can we be more quantitative?

$\Downarrow$

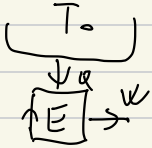
Clausius: $\sum_i \frac{\delta Q_i}{T_i} \leq 0$

or $\oint \frac{dQ}{T_{ext}} \leq 0$

Proof: Use Carnot engines to replenish the heat from a common heat bath at T_0



forms one big engine:



$\Delta U = W + Q = 0$

$W = \sum_i \delta W_i \quad Q = \sum_i \delta Q_i \frac{T_0}{T_i}$

2nd law $\Rightarrow Q = -W \leq 0$

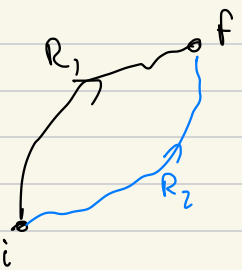
$\Rightarrow \sum_i \frac{\delta Q_i}{T_i} \leq 0$

If process is <u>reversible</u> $\Rightarrow T_{ext} = T_{system} = T$

we can run the process backwards:

$$\oint_{rev.} \frac{dQ}{T} \leq 0 \qquad \oint_{rev} \frac{dQ}{T} \geq 0 \Rightarrow \oint_{rev.} \frac{dQ}{T} = 0$$

Two reversible processes between same endpoints:



$$\int_{R_1} \frac{dQ}{T} - \int_{R_2} \frac{dQ}{T} = \oint \frac{dQ}{T} = 0$$

$$\Rightarrow \int_{rev.} \frac{dQ}{T} \text{ is independent of path!}$$

$$\Rightarrow \int_A^B \frac{dQ_{rev.}}{T} = S_f - S_i \equiv \Delta S \quad \text{for some } S(V, P, \ldots)$$

↑ entropy

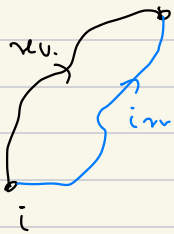
infinitesimally: $dQ_{rev.} = T\,dS$ (compare: $dW = -P\,dV$)

$$S(P,V,\ldots) = \int_{(P_0,V_0,\ldots)}^{(P,V,\ldots)} \frac{dQ_{rev}}{T}$$

we'll later fix reference point to be $T = 0$ (3rd law)

<u>usually</u> S is extensive (i.e. if U and V are both additive)

More generally, compare a reversible path with an arbitrary one



$$\int_{\text{irrev}} \frac{dQ}{T_{ext}} - \underbrace{\int_{\text{rev.}} \frac{dQ}{T}}_{\Delta S} = \oint \frac{dQ}{T_{ext}} \leq 0$$

$$\Rightarrow S_f - S_i \geq \int_i^f \frac{dQ}{T_{ext}} \qquad \text{or} \qquad dS \geq \frac{dQ}{T_{ext}}$$

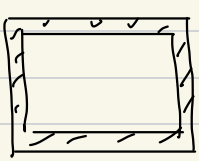
In an adiabatic process: $dQ = 0$

$$\Rightarrow \boxed{\Delta S \geq 0}$$ (Planck's version of 2nd law)

$\Delta S = 0 \Leftrightarrow$ reversible process

Fully isolated system: $W = 0$

$\rightarrow U = \text{const.}$

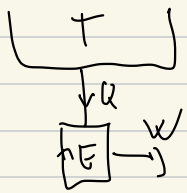


Equilibrium: S is maximal at fixed U

note: entropy of a <u>subsystem</u> could increase

Recovering other forms of 2nd law:

Kelvin:



system → $\Delta S_{tot} = \Delta S_{sys} + \Delta S_{env}$ ← bath

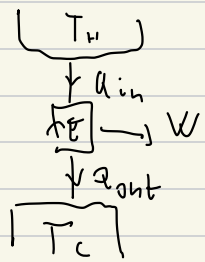
$= 0 - \frac{Q}{T} < 0$ ✗

cyclic!

real engine:

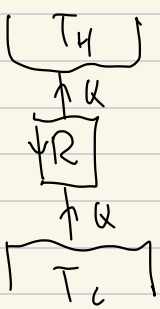
$\Delta S_{tot} = \Delta S_H + \Delta S_C$

$\simeq -\frac{Q_{in}}{T_H} + \frac{Q_{out}}{T_C}$



reversible: $\Delta S_{tot} = 0 \rightsquigarrow \frac{Q_{out}}{Q_{in}} = \frac{T_C}{T_H}$ Carnot

Clausius:



$\Delta S_{tot} \simeq \Delta S_H + \Delta S_C \simeq$

$\simeq -\frac{Q}{T_C} + \frac{Q}{T_H}$

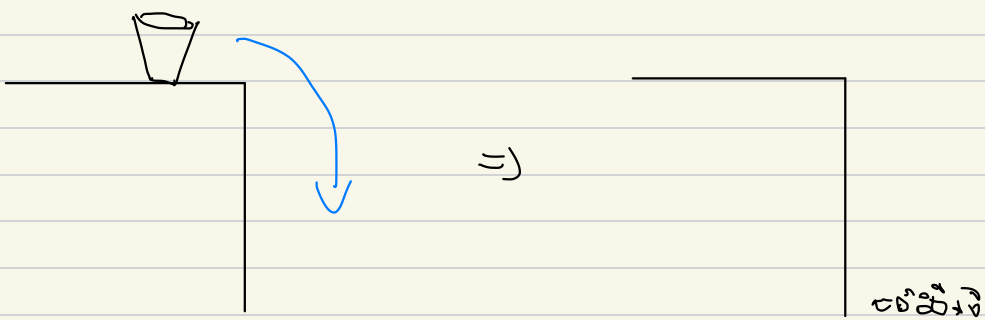
$\simeq Q\left(\frac{1}{T_H} - \frac{1}{T_C}\right) < 0$ ✗

Irreversibility: $S_f > S_i$ reverse process cannot happen

→ <u>arrow of time</u>

Where does it come from if microscopic laws (e.g. Newton) are reversible?

Paradigmatic example: glass shattering



Many different ways of arranging the shards that look very similar, but just one way of putting them back together

→ entropy counts the number of different configurations that "look the same"

→ statistical physics

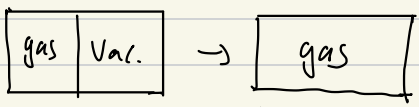
Examples:

(1) Heating at constant P; want $S(T_f, P) - S(T_i, P)$

if reversible: $dQ = C_p\, dT \rightarrow \Delta S = \int_i^f C_p \frac{dT}{T}$

assuming $C_p(T)$ constant $\rightarrow \Delta S = C_p \ln\left(\frac{T_f}{T_i}\right)$

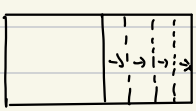
(2) Free expansion of ideal gas:



V_i V_f

$T = \text{const}$; $Q = 0$

irreversible → consider a reversible process with same endpoints



ideal gas: $U = U(T) = \text{const.}$

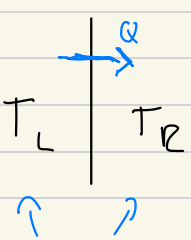
$\rightarrow dQ = -dW = P\,dV$

$$\Delta S = \int \frac{dQ}{T} = \int \frac{P\,dV}{T} = \int nR \frac{dV}{V} = nR \ln\left(\frac{V_f}{V_i}\right)$$

If $V_f = 2V_i \rightarrow \Delta S = nR \ln 2$ indep. of T, V

↳ 2 options per particle

(3) Heat flow



both infinite $T_L, T_R = \text{const.}$

$$\Delta S = \Delta S_L + \Delta S_R = \frac{-Q}{T_L} + \frac{Q}{T_R} = Q\left(\frac{1}{T_R} - \frac{1}{T_L}\right)$$

$\Delta S > 0 \iff Q > 0$ and $T_L > T_R$
or $Q < 0$ and $T_L < T_R$

→ heat flows from hot to cold

T-S diagrams:

Instead of (P, V) we can represent a system's state with the pair (T, S)

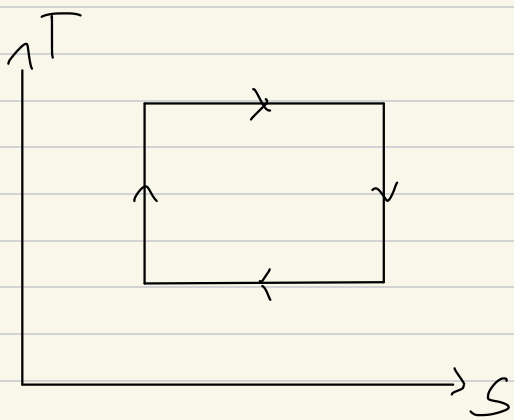
Isotherm: $T = \text{const.}$

reversible adiabatic process: $S = \text{const.}$

$(dS = T\,đQ_{rev} = 0)$

↓

Carnot process:



for cyclic process on T-S diagram

$$Q = \oint T\,dS = \text{area enclosed}$$

Fundamental thermodynamic relation

Combine 1st and 2nd law:

$$dU = dQ + dW = -PdV + TdS$$

$$\hookrightarrow \Delta U = \int_{\text{initial}}^{\text{final}} (-PdV + TdS)$$

U, S, V are <u>all</u> state variables

↳ only depend on initial / final point

→ $dU = -PdV + TdS$ also holds for non-reversible processes

If other types of work are also involved:

$$dU = TdS - PdV + \mathcal{E}dz + E\,dP_{el} + B\,dM + \dots$$

Example: entropy of ideal gas

$$U = U(T) \rightarrow dU = c_V\, dT$$

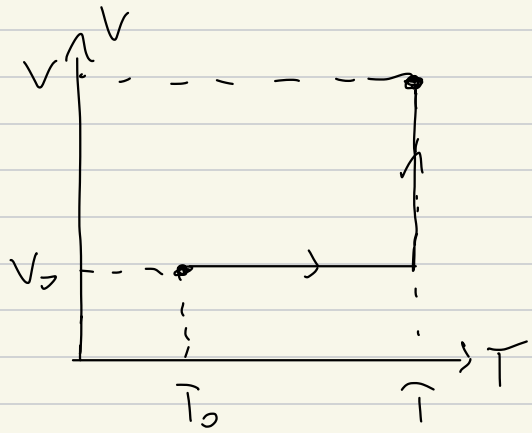
$$T\,dS = c_V\, dT + P\,dV = c_V\, dT + nRT\,\frac{dV}{V}$$

$$PV = nRT$$

$$dS = c_V\,\frac{dT}{T} + nR\,\frac{dV}{V}$$

$$\Delta S = S - S_0 = \int_{(T_0,V_0)}^{(T,V)} dS =$$

$$= c_V \ln T + nR \ln V$$



Thermodynamic potentials

fundamental relation: $dU = TdS - PdV$ (1)

If we write $U = U(S,V)$ ("natural" variables)

↳ $dU = \frac{\partial U}{\partial S} dS + \frac{\partial U}{\partial V} dV$ (2)

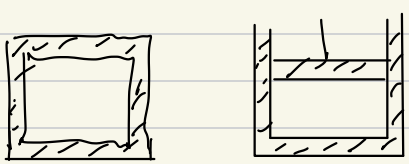
(1) + (2) ⇒ $T = \left(\frac{\partial U}{\partial S}\right)_V$; $P = -\left(\frac{\partial U}{\partial V}\right)_S$

A consequence: $\left(\frac{\partial T}{\partial V}\right)_S = \frac{\partial^2 U}{\partial V \partial S} = -\left(\frac{\partial P}{\partial S}\right)_V$

(Maxwell relation)

Reminder: in equilibrium $S = \max$ at constant U, V

What if $V \neq$ const.?



V = const. $V \neq$ Const.

If P = const: $Q = \Delta U - W = \Delta U + P\Delta V = \Delta(U + PV) = \Delta H = 0$

In fact if $Q = 0 \rightarrow W$ is independent of path

enthalpy

↳ if $P_i = P_f = P$: $\Delta U = W = -P\Delta V$ and $\Delta H = 0$

equilibrium: $S = \max$ at constant H

More generally, if P is constant: $Q = \Delta H$

↳ relevant e.g. to chemical reactions

Most general case: $dH = dU + PdV + Vdp \simeq TdS + Vdp$

natural variables: $H = H(S, P)$

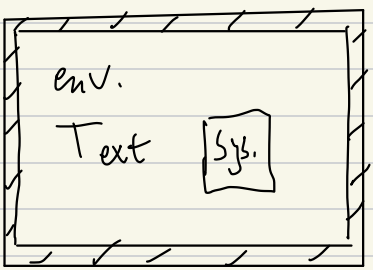
(exchanged $V \leftrightarrow P$ compared to U)

$$\rightarrow T = \left(\frac{\partial H}{\partial S}\right)_P \; ; \quad V = \left(\frac{\partial H}{\partial P}\right)_S$$

$$\rightarrow \left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

We considered equilibrium conditions for systems enclosed in adiabatic walls ($Q=0$)

More realistic: system in an environment at fixed T



what replaces $S = \max.$?

Isothermal (T = const.) reversible process:

$$W = \Delta U - Q = \Delta U - T\Delta S = \Delta(U - TS) \equiv \Delta F$$

$F = U - TS$ (Helmholtz) free energy

More generally: let $T_i = T_f = T_{ext}$ and V = const.

system + environment together are closed $\Rightarrow \Delta S_{sys} + \Delta S_{env.} \geq 0$

$$dU_{env} = T_{ext}\, dS_{env} \Rightarrow \Delta S_{env} = \frac{1}{T_{ext}} \Delta U_{env}$$

energy conservation: $\Delta U_{sys} + \Delta U_{env} - W = 0$

(reminder: $-W$ is work done by system)

$$\Rightarrow \Delta S_{sys} + \frac{1}{T_{ext}}(W - \Delta U_{sys}) \geq 0$$

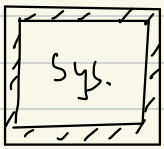
$$\Rightarrow -W \leq -\Delta(U - TS)_{sys} = -\Delta F_{sys}$$

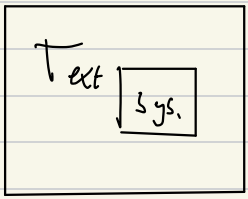
Most work we can get out

Special case: $W = 0$

$\rightarrow \Delta F \leq 0$ (similar to $\Delta S \geq 0$ when $Q = 0$)

$\Rightarrow$ In equilibrium at constant V, T: $F = \underline{\min}$





$S = \max$ at $U, V = \text{const}$. $F = \min$. at $T, V = \text{const}$.

we will see that $F = \min$. amounts to $T = T_{ext}$

$F = U - TS = \min$.

At $T = 0$: $U = \min$ ("ground state")

At $T \to \infty$: $S = \max$. (no constraint on U!)

E.g.: melting of ice.

Solid crystal structure has lower U and S than liquid

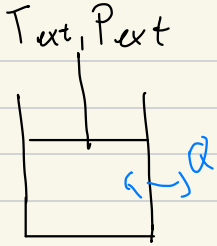
low T: solid preferred; high T: liquid / gas

$$dF = dU - TdS - SdT = -SdT - PdV$$

natural variables: $F = F(T, V)$

$$\rightarrow S = -\left(\frac{\partial F}{\partial T}\right)_V \; ; \quad P = -\left(\frac{\partial F}{\partial V}\right)_T$$

$$\rightarrow \frac{\partial S}{\partial V} = \frac{\partial P}{\partial T}$$



What if $V \neq$ const. and $Q \neq 0$?

If $P, T =$ const. reversible process:

$$\Delta U = Q + W = T\Delta S - P\Delta V + W_{useful}$$

$P\Delta V$: work done by environment

W_{useful}: additional useful work done by the system

$$\Rightarrow W_{useful} = \Delta(U - TS + PV) = \Delta G$$

$G = U - TS + PV$: Gibbs free energy

More generally: $T_i = T_f = T_{ext}$; $P_i = P_f = P_{ext}$

$$\Delta S_{sys} + \Delta S_{env} \geq 0$$

$$dU = TdS - PdV \rightarrow \Delta S_{env} = \frac{1}{T_{ext}}(\Delta U_{env} + P_{ext}\Delta V_{env})$$

$$\Delta U_{env} = -\Delta U_{sys} + W_{useful} \; ; \; \Delta V_{env} = -\Delta V_{sys}$$

$$\rightarrow -W_{useful} \leq T_{ext}\Delta S_{sys} - \Delta U_{sys} - P_{ext}\Delta V_{sys} = -\Delta G_{sys}$$

$$W_{useful} = 0 \rightarrow \Delta G \leq 0$$

↳ in equilibrium: $\Delta G = \min.$ (at $T, P =$ const.)

$$dG = dU - TdS - SdT + PdV + VdP$$

$$= -SdT + VdP$$

natural variables: $G = G(T, P)$

$$\rightarrow S = -\left(\frac{\partial G}{\partial T}\right)_P \quad ; \quad V = \left(\frac{\partial G}{\partial P}\right)_T$$

$$\rightarrow \left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

G is important in many applications where T, P are constant.

E.g.: chemical reactions

If $\Delta G_{reac.} < 0$: happens spontaneously

$\Delta G_{reac.} > 0$: needs energy (e.g. from sunlight)

note: if $T = const. \rightarrow \Delta G = \Delta H - T\Delta S$

→ more reactions can happen spontaneously at higher T

Thermo. pot.	Nat. Var.	Thermo. rel.	Maxwell rel.
U	V, S	$dU = TdS - PdV$	$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$
$H = U + PV$	P, S	$dH = TdS + VdP$	$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$
$F = U - TS$	V, T	$dF = -SdT - PdV$	$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$
$G = U + PV - TS$	P, T	$dG = -SdT + VdP$	$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$

Application of Maxwell relations: energy of ideal gas

$$dU = TdS - PdV \rightarrow \left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - P$$

Maxwell $\rightarrow$ $\left(\frac{\partial U}{\partial V}\right)_T = T\underbrace{\left(\frac{\partial P}{\partial T}\right)_V} - P$ ("energy equation")

↑ more measurable than $\frac{\partial S}{\partial V}$

Ideal gas: $PV = nRT \rightarrow \left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V} = \frac{P}{T}$

$\rightarrow \left(\frac{\partial U}{\partial V}\right)_T = 0 \rightarrow U = U(T)$ independent of V

"Cyclic rule":

$f(x,y,z) = 0$ (e.g. equation of state)

↳ any two variables of x, y, z determine the third

$$dx = \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y dz$$

$$dy = \left(\frac{\partial y}{\partial x}\right)_z dx + \left(\frac{\partial y}{\partial z}\right)_x dz$$

$$\rightarrow dx = \underbrace{\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial x}\right)_z}_{①} dx + \underbrace{\left[\left(\frac{\partial x}{\partial z}\right)_y + \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x\right]}_{②} dz$$

Choose x and z as the two independent variables

↳ consistency requires: ① $= 1$ "reciprocal rule"

② $= 0$ "cyclic rule"

Relationships between c_p and c_V:

$$C = \frac{dQ}{dT} = T\frac{\partial S}{\partial T} \quad \text{(for reversible processes)}$$

$$c_p = T\left(\frac{\partial S}{\partial T}\right)_p = T\left(\frac{\partial S}{\partial T}\right)_V + T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p$$

$S = S(V, T)$ — $\left(\frac{\partial S}{\partial T}\right)_V$: C_V — $\left(\frac{\partial S}{\partial V}\right)_T$: $\left(\frac{\partial P}{\partial T}\right)_V$ (Maxwell rel.)

cyclic rule: $\left(\frac{\partial P}{\partial T}\right)_V = -\left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$

$$\left(\frac{\partial V}{\partial T}\right)_p = V \cdot \beta \quad \text{(thermal expansion)}$$

$$\left(\frac{\partial P}{\partial V}\right)_T = \left(\frac{\partial V}{\partial P}\right)_T^{-1} = \frac{-1}{V\kappa_T} \quad \text{(isothermal compressibility)}$$

$$\Rightarrow \boxed{c_p = c_V + T\frac{\beta^2}{\kappa_T}V} > c_V$$

Usually c_V is easier to calculate theoretically while c_p is easier to measure in experiments

Ratio: $\frac{C_P}{C_V} = \frac{\left(\frac{\partial S}{\partial T}\right)_P}{\left(\frac{\partial S}{\partial T}\right)_V} \overset{\text{cyclic rule}}{=} \frac{-\left(\frac{\partial S}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_S}{-\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_S} =$

$= \frac{\left(\frac{\partial V}{\partial S}\right)_T \left(\frac{\partial S}{\partial P}\right)_T}{\left(\frac{\partial V}{\partial T}\right)_S \left(\frac{\partial T}{\partial P}\right)_S} \overset{\text{reciprocal rule}}{=} \frac{\left(\frac{\partial V}{\partial P}\right)_T}{\left(\frac{\partial V}{\partial T}\right)_S} = \frac{\kappa_T}{\kappa_S}$

chain rule

"isenthalpic" compressibility

Similar relationship holds for other pairs of extensive-intensive variables

e.g. magnetization M and magnetic field B

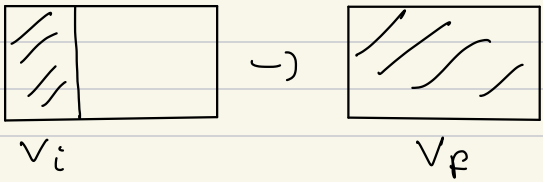
magnetic susceptibility: $\chi = \frac{1}{V}\frac{\partial M}{\partial B}$

(measures how difficult it is to magnetize material)

$\rightarrow \frac{\chi_T}{\chi_S} = \frac{\left(\frac{\partial M}{\partial B}\right)_T}{\left(\frac{\partial M}{\partial B}\right)_S} = \frac{C_B}{C_M} = \frac{T\left(\frac{\partial S}{\partial T}\right)_B}{T\left(\frac{\partial S}{\partial T}\right)_M}$

Joule effect

free expansion: ($W = 0$)



$$\Delta T = T_f - T_i = T(V_f, U_f) - T(V_i, U_i)$$

If adiabatic (e.g. happens quickly): $Q = 0 \rightarrow U_f = U_i$

↳ $\Delta T = T(V_f, U) - T(V_i, U)$

Replace it with a quasi-static process between same endpoints (and U = const.)

↳ $$\Delta T = \int_{V_i}^{V_f} dV \left(\frac{\partial T}{\partial V}\right)_U$$

$\left(\frac{\partial T}{\partial V}\right)_U = \mu_J$ (Joule coefficient)

(cyclic)

$$\mu_J = -\left(\frac{\partial T}{\partial U}\right)_V \left(\frac{\partial U}{\partial V}\right)_T = \frac{1}{C_V}\left(P - T\left(\frac{\partial P}{\partial T}\right)_V\right)$$

$\left(\frac{\partial T}{\partial U}\right)_V = 1/C_V$

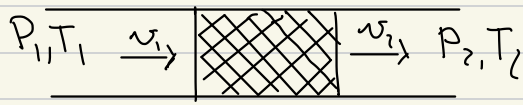
$\left(\frac{\partial U}{\partial V}\right)_T$ = energy equation

ideal gas: $T\left(\frac{\partial P}{\partial T}\right)_V = P \rightarrow \mu_J = 0$

real gases: $\mu_J < 0$ (attraction between particles ↳ U grows with V)

Joule-Thomson effect

Throttling process: fluid flowing through valve/plug

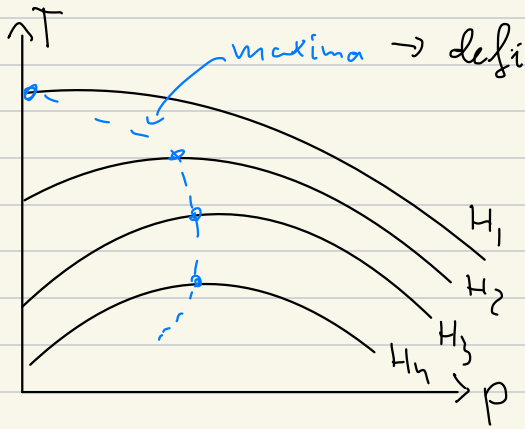


$p_2 < p_1$

We saw earlier (p. ??): $v_1^2 - v_2^2 = 2(H_2 - H_1)$

steady flow: $v_1 \approx v_2 \rightarrow H_1 = H_2$ isenthalpic process

Isenthalps (H = const. curves): ← fix p_1, T_1, p_2 and measure T_2



above: $T_2 > T_1$ (heating)

below: $T_2 < T_1$ (cooling)

↳ can be used to refrigerate (e.g. Linde liquification)

Replace irreversible throttling with reversible process: (same endpoints, H = const.)

$$\Delta T = \int_{p_1}^{p_2} \left(\frac{\partial T}{\partial p}\right)_H dp$$

$\left(\frac{\partial T}{\partial p}\right)_H =: \mu_{JK}$ Joule-Kelvin coeff.

cyclic

$$\mu_{JK} = -\left(\frac{\partial T}{\partial H}\right)_P \left(\frac{\partial H}{\partial P}\right)_T$$

$\left(\frac{\partial T}{\partial H}\right)_P = 1/C_P$

$$dH = TdS + VdP \rightarrow \left(\frac{\partial H}{\partial P}\right)_T = T\left(\frac{\partial S}{\partial P}\right)_T + V$$

$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$ (Maxwell)

$$\mu_{JK} = \frac{1}{C_P}\left(T\left(\frac{\partial V}{\partial T}\right)_P - V\right)$$

↑ RHS determined by eq. of state

Ideal gas: $T\frac{\partial V}{\partial T} = V \rightarrow \mu_{JK} = 0$

Open systems and chemical potential

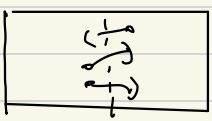
So far we only considered processes where particle number N is constant

Not always true!

E.g. -) Chemical reactions

-) Melting / evaporation

-) Porous boundaries:



(particles can diffuse through)

When a particle leaves / enters it takes some energy with it

↳ $dU = TdS - PdV + \mu dN$

$U = U(S, V, N)$

$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}$: chemical potential

Other potentials also change:

$dF = -SdT - PdV + \mu dN \rightarrow \mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$

$dG = -SdT + VdP + \mu dN \rightarrow \mu = \left(\frac{\partial G}{\partial N}\right)_{T,P}$ ← better for measuring μ

$G = G(T, P, N)$ only G and P are extensive

$\rightarrow G = N\,\phi(T, P) \rightarrow \mu = \left(\frac{\partial G}{\partial N}\right)_{T,P} = \phi(T, N) = \frac{G}{N}$

("Gibbs free energy / particle")

Multiple particle species: $N_1, N_2, N_3, \ldots$

$U = U(S, V, N_1, N_2, \ldots)$

$$dU = T\,dS - P\,dV + \sum_i \mu_i\, dN_i$$

where $\mu_i = \left(\frac{\partial U}{\partial N_i}\right)_{S, V, N_{j \neq i}}$

Euler's thm: let $f(x_1, x_2, \ldots)$ be a homogenous f.,

$$f(\alpha x_1, \alpha x_2, \ldots) = \alpha\, f(x_1, x_2, \ldots)$$

$$\Rightarrow f(x_1, x_2, \ldots) = \sum_i x_i \frac{\partial f}{\partial x_i}$$

We can apply this to $U(S, V, N_1, N_2, \ldots)$
since all variables appearing are extensive:

$$\rightarrow U = S\frac{\partial U}{\partial S} + V\frac{\partial U}{\partial V} + \sum_i N_i \frac{\partial U}{\partial N_i} = TS - VP + \sum_i \mu_i N_i$$

$$\Rightarrow G = U - TS + PV = \sum_i \mu_i N_i$$

Example application: chemical reactions

e.g. $H_2 + Cl_2 \rightleftharpoons 2\,HCl$

Eventually system settles down to <u>chemical equilibrium</u>

↳ reaction happens in both directions at same rate

equilibrium: $G = \min. \rightarrow dG = 0$

$P, T = \text{const.} \rightarrow dG = \sum_i \mu_i \, dN_i$

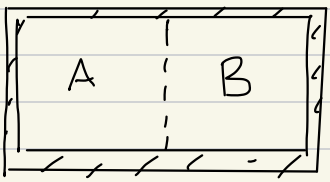
→ chemical equilibrium $\Leftrightarrow$ $\sum_i \mu_i \, dN_i = 0$

↑ this fixes the ratios of μ_i

e.g. in above example $d\,HCl = -2\,d\,H_2 = -2\,d\,Cl_2$

↳ $2\mu_{HCl} = \mu_{H_2} + \mu_{Cl_2}$ in equilibrium

General equilibrium condition



A and B can exchange energy and particles

Wall can move

$U_A + U_B$, $V_A + V_B$, $N_A + N_B$ are constant

$S_{tot} = S_A(U_A, V_A, N_A) + S_B(U_B, V_B, N_B)$

Equilibrium: $S_{tot} = max. \rightarrow dS_{tot} = 0$

$$\hookrightarrow 0 = dS_A + dS_B = \frac{1}{T_A}(dU_A + P_A dV_A - \mu_A dN_A) + \frac{1}{T_B}(dU_B + P_B dV_B - \mu_B dN_B)$$

$$= dU_A\left(\frac{1}{T_A} - \frac{1}{T_B}\right) + dV_A\left(\frac{P_A}{T_A} - \frac{P_B}{T_B}\right) - dN_A\left(\frac{\mu_A}{T_A} - \frac{\mu_B}{T_B}\right)$$

Needs to be true for any dU_A, dV_A, dN_A

$$\rightarrow \boxed{T_A = T_B \; ; \; P_A = P_B \; ; \; \mu_A = \mu_B}$$

If $T_A = T_B$; $P_A = P_B$ but $\mu_A \neq \mu_B$: $dS = \frac{dN_A}{T}(\mu_B - \mu_A)$

$dS \geq 0 \rightarrow$ particles flow from high to low μ

If V_A, N_A fixed: $dS = dU_A\left(\frac{1}{T_A} - \frac{1}{T_B}\right) \rightarrow T_A \overset{!}{=} T_B$

3rd law

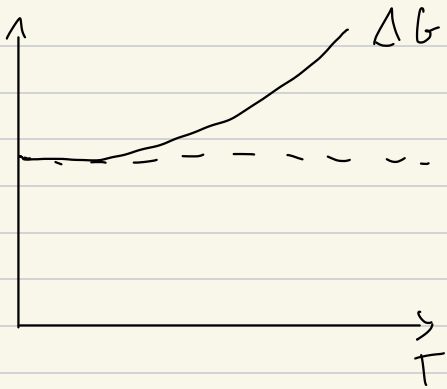
Using 2nd law, we found a universal ("thermodynamic") temperature scale

↳ only meaningful for $T \geq 0$ ("absolute zero")

What happens as $T \to 0$?

Nernst experiment on chemical reactions:

measure ΔG as a function of T



← becomes horizontal:

$$\lim_{T \to 0} \frac{\partial}{\partial T}(\Delta G) = 0$$

$$S = -\left(\frac{\partial G}{\partial T}\right)_p \to \frac{\partial}{\partial T}(\Delta G) = \Delta S$$

"Nernst heat theorem": For any process between two states at the same temperature T

($\equiv$ 3rd law)

$$\lim_{T \to 0} \Delta S = 0$$

Planck's formulation: $\lim_{T \to 0} S = S_0$ is same for all substances

So far we only defined entropy up to an additive constant

→ We can choose $S_0 = 0$

This amounts to using $T = 0$ state as the reference point in defining entropy

$$S = \int_0^T \frac{dQ_{rev}}{T}$$

Note: 3rd law is not as universal as 0, 1, 2 laws

It can fail for certain special systems

↳ "residual entropy" $\neq 0$

A somewhat more general version: $\lim_{T \to 0} \frac{S}{N} = s_0 = 0$

($\frac{S}{N}$: entropy density)

But might still fail e.g. for glasses

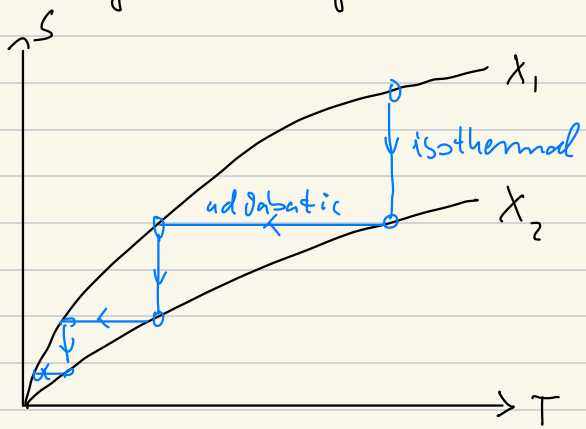
↳ residual entropy density

Consequences:

(1) Unattainability of $T = 0$ (in finite number of steps)

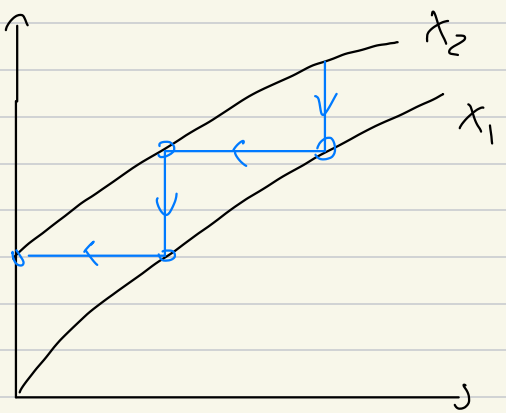
Illustration: imagine we try to approach $T = 0$ by switching between two values of some parameter X

e.g. pressure, volume, magnetic field, ...



← T only approaches 0 asymptotically (needs ∞ no. of steps)

If 3rd law failed:



← converges after a finite no. of steps

Note: This limits the efficiency of a Carnot engine

$\eta = 1 - \frac{T_C}{T_H} < 1$ Cannot maintain $T_C = 0$

With equations:

$$S(T_1, X_1) = \underbrace{S(0, X_1)}_{= S_0} + \int_0^{T_1} dT \left(\frac{\partial S}{\partial T}\right)_{X = X_1}$$

$$S(T_2, X_2) = \underbrace{S(0, X_2)}_{= S_0} + \int_0^{T_2} dT \left(\frac{\partial S}{\partial T}\right)_{X = X_2}$$

consider one of the adiabatic steps

↳ most efficient for cooling (otherwise get heat leaking in)

If we also make it reversible: $\Delta S = 0$

$$\Delta S = \int_0^{T_2} dT \left(\frac{\partial S}{\partial T}\right)_{X = X_2} - \int_0^{T_1} dT \left(\frac{\partial S}{\partial T}\right)_{X = X_1} = 0$$

want to end up in $T_2 = 0$

$$\rightarrow \int_0^{T_1} dT \left(\frac{\partial S}{\partial T}\right)_{X = X_1} = 0$$

but at $T > 0$: $\frac{\partial S}{\partial T} > 0$ → inconsistent

< 0: instability ; $= 0$: temperature makes no sense

($C_V = T \frac{\partial S}{\partial T}$ ⇐ no heat exchange with env.)

(2) Heat capacity

$$C_V = T\frac{\partial S}{\partial T} \rightarrow S - S_0 = \int_0^T dT' \frac{C_V}{T'}$$

note: this is how 3rd law

$$\lim_{T\to 0} S = S_0 \Rightarrow \lim_{T\to 0} C_V = 0$$

(3) Thermal expansion:

$$\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P \overset{\text{Maxwell}}{=} -\frac{1}{V}\left(\frac{\partial S}{\partial P}\right)_T$$

$$\Rightarrow \lim_{T\to 0} \beta = 0$$

Statical Physics

System of N particles, $N \gtrsim 10^{23}$

↳ impossible to describe exactly

Also unnecessary: too much information

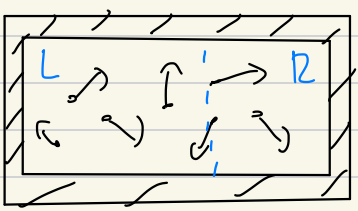
Instead of treating particles individually ("microstate")

focus on properties of the whole ("macrostate")

Since we don't know what each particle is doing,

we will need a <u>probabilistic</u> description

Illustrative example: gas in a box



Each particle is either on the left (L) or right (R)

Microstate: specify L or R for each particle ⇒ 2^N options

↳ <u>exponentially</u> many in N

Macrostate: Number of particles on L/R: $N_L, N_R = N - N_L$

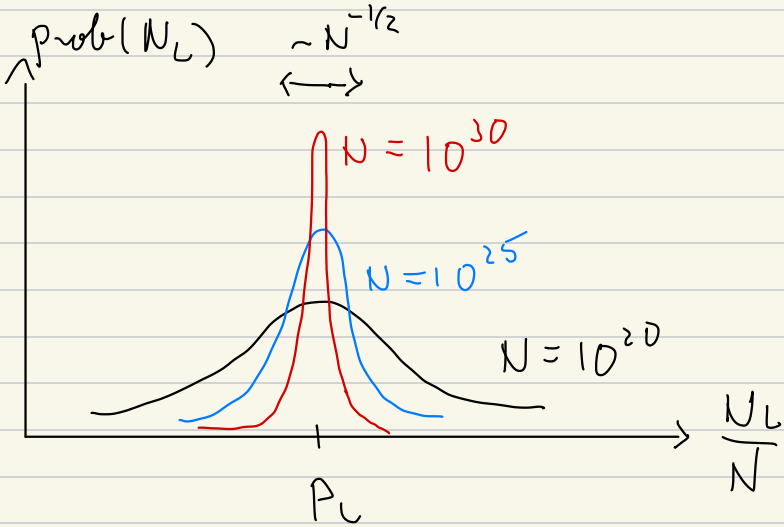
↳ density N_L/N is just one number

Key point: macroscopic properties become deterministic as $N \to \infty$

E.g. let each particle be on the left / right with probabilities p_L, $p_R = 1 - p_L$ (independent for each particle)

$$\text{Prob}(N_L \text{ particles on left}) = \binom{N}{N_L} p_L^{N_L} p_R^{N_R}$$

Average: $\bar{N}_L = \sum_{N_L=0}^{N} \text{prob}(N_L)\, N_L = p_L \cdot N$



Fluctuations in $\frac{N_L}{N}$ go to 0 as $N \to \infty$

N particles (atoms / molecules)

We will treat them as classical → Newton's laws

note: at low temperatures, quantum mechanics becomes important (see later)

Full microscopic description:

N positions $\underline{x}_i$ $\quad i = 1, \ldots, N$

N velocities/momenta: $\underline{v}_i$ or $\underline{p}_i = \underline{v}_i / m_i$

6N parameters → phase space $\{\underline{x}_i, \underline{p}_i\}_{i=1}^{N}$

note: multi-atom molecules: additional rotational and vibrational degrees of freedom

2-atom: 

3 atoms: 

Total energy:

$$E(\underline{x}_1, \ldots, \underline{x}_N, \underline{p}_1, \ldots, \underline{p}_N) = \underbrace{\sum_i \frac{p_i^2}{2m_i}}_{\text{kinetic}} + \underbrace{\sum_i V(\underline{x}_i)}_{\text{potential}} + \underbrace{\frac{1}{2}\sum_{i,j} u(\underline{x}_i - \underline{x}_j)}_{\text{interaction}}$$

system at rest: $\sum_i \underline{p}_i = 0$

closed system: $V(\underline{x}) = 0$

} we can equate $E \equiv U$ (internal energy)

Fix N, V, U: Phase space is finite

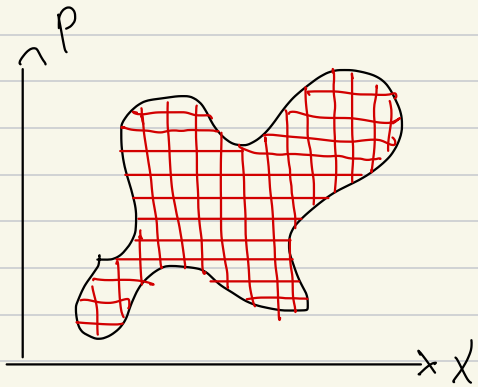


x, p can only be measured with some finite precision $\Delta x, \Delta p$

fundamental limit from quantum mechanics: $\Delta x \Delta p \geq \frac{\hbar}{2}$

→ phase space is divided into "cells" ≃ microstates

$\uparrow$
ω



Need to define macrostates

↳ depends on what we can/want to measure

E.g. divide system into regions with volumes V_m

could be just left/right or more

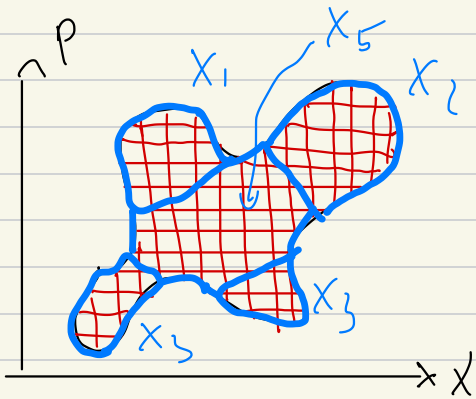
densities: $n_m = N_m / V_m$; $\epsilon_m = U_m / V_m$

↳ macrostates X

Regions contain many particles

↳ each macrostate X corresponds to many microstates

→ phase space "volume" $\Omega(X)$



macrostates $\sim N^0$ (assumption)

microstates $\sim e^N$ (e.g. $(V/\Delta x^3)^N$ positions)

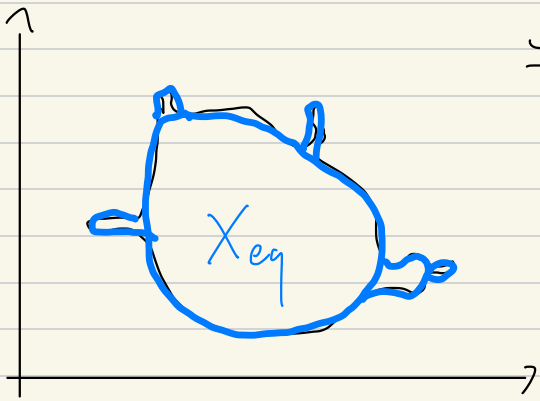
↳ most $\Omega \sim e^N$

There is one particular macrostate that corresponds to the system being in equilibrium

Fundamental assumption: $\Omega(X)$ is maximal when $X = X_{eq}$.

Why? → because largest macrostate contains almost all microstates

More accurate drawing:



$$\frac{\Omega(X_{eq})}{\Omega_{tot}} = 1 - \varepsilon$$

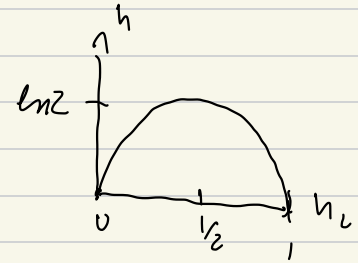
$$\varepsilon \sim e^{-O(N)}$$

Starting from some other macrostate, we quickly fall into X_{eq} and never get out

E.g. let macrostate be specified by $n_L = N_L/N$

$\binom{N}{N_L} = \binom{N}{N n_L}$ possibilities / microstates

claim: $\binom{N}{N n_L} \sim e^{N h(n_L)}$



Stirling approx: $N! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot N$

$$\rightarrow \ln(N!) = \ln(1) + \ln(2) + \ldots + \ln(N)$$

$$= \sum_{j=1}^{N} \ln(j) \approx \int_1^N \ln x \, dx = \left[x \ln x - x \right]_1^N$$

$$\approx N \ln N - N \approx N \ln N$$

$$\ln \binom{N}{N n_L} = \ln N! - \ln(N n_L)! - \ln(N(1-n_L))! \approx$$

$$\approx N \ln N - N \cdot n_L \cdot \ln(N n_L) - N(1-n_L) \ln(N(1-n_L))$$

$$= N \ln N - N n_L \ln N - N n_L \ln n_L$$

$$- N(1-n_L) \ln N - N(1-n_L) \ln(1-n_L)$$

$$= N \left(-n_L \ln n_L - (1-n_L) \ln(1-n_L) \right)$$

Equilibration:



$\Omega(X)$ grows and becomes maximal in eq.

$\Updownarrow$

Therm.: S grows and becomes maximal

S: extensive $\sim N$

$\Omega \sim e^{O(N)}$

$\Big\}$ $S_{\text{Boltzmann}}(X) = k_B \ln \Omega(X)$

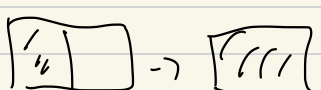
$k_B \simeq 1.381 \cdot 10^{-23} \frac{J}{K}$

$\downarrow$

S is "statistical": only defined for macrostate not micro

↳ this is why it can behave irreversibly

growth of S: probabilistic statement

e.g. free expansion:  $\Omega \to 2^N \Omega$

$S \to S + N \ln 2$

In equilibrium: $\ln \Omega(X_{eq}) \approx \ln \Omega_{tot}(U, V, N)$

$\rightarrow S = S(U, V, N)$

$$\frac{1}{T} := \left(\frac{\partial S}{\partial U}\right)_{V,N} \qquad p := T\left(\frac{\partial S}{\partial V}\right)_{U,N}$$

$$\mu := -T\left(\frac{\partial S}{\partial N}\right)_{U,V}$$

Boltzmann approach: follow macrostate

Gibbs approach: Assign probabilities p_w to microstates w

If we only fix macrostate (X): natural to assign same p_w to all w consistent with it

$$p_w = \begin{cases} 0 & \text{if } w \notin X \\ \frac{1}{\Omega(X)} & \text{if } w \in X \end{cases}$$

In equilibrium: $\Omega(X_{eq}) \to \Omega_{tot}(U,V,N)$

Fix V,N $\to$ $\Omega_{tot} = \Omega_{tot}(U)$ ("energy shells")

$\hookrightarrow$ $$p_w = \begin{cases} 0 & E_w \neq U \\ \frac{1}{\Omega_{tot}(U)} & E_w = U \end{cases}$$

In equilibrium: all allowed states are equally likely

$\hookrightarrow$ microcanonical ensemble: $p_w^{m.c.}$ measures uncertainty

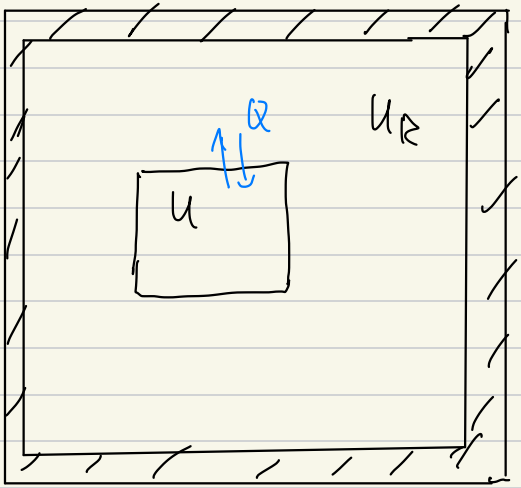
Also need to redefine entropy: $S_{Gibbs}(\{p_w\}) = -k_B \sum_w p_w \ln p_w$

In equilibrium: $S_G[p_w^{m.c.}] = -k_B \sum_{w: E_w = U} \frac{1}{\Omega_{tot}(U)} \ln \frac{1}{\Omega_{tot}(U)}$

$= k_B \ln \Omega_{tot}(U) = S_{Boltzmann}$

↑ maximizes S_G over all p_w (with fixed U,V,N)

Gibbs approach becomes necessary once we consider systems that are not isolated but in contact with a reservoir (bath)



$U_{tot} = U + U_R$ fixed

Fix microstate w of sys.

↳ $U = E_w$

$\Omega_R(U_R) = \Omega_R(U_{tot} - E_w)$

$\Omega_{tot} = \sum_w \Omega_R(U_{tot} - E_w)$

S+R is closed → apply microcanonical ensemble:

$$P_w = \frac{\Omega_R(U_{tot} - E_w)}{\sum_w \Omega_R(U_{tot} - E_w)}$$

bath >> system
$U_{tot} >> E_w$

$1/T_R$

$$\Omega_R(U - E_w) = e^{S_R(U_{tot} - E_w)/k_B} \approx e^{S(U_{tot})/k_B} e^{-\frac{E_w}{k_B}\frac{\partial S_R}{\partial U_R}}$$

$$\rightarrow \boxed{P_w^{can} = \frac{e^{-E_w/k_BT}}{\sum_w e^{-E_w/k_BT}}}$$

canonical ensemble
(in equilibrium: $T = T_R$)

Partition fct. $Z = \sum_w e^{-E_w/k_BT}$

Energy is now not fixed but probabilistic

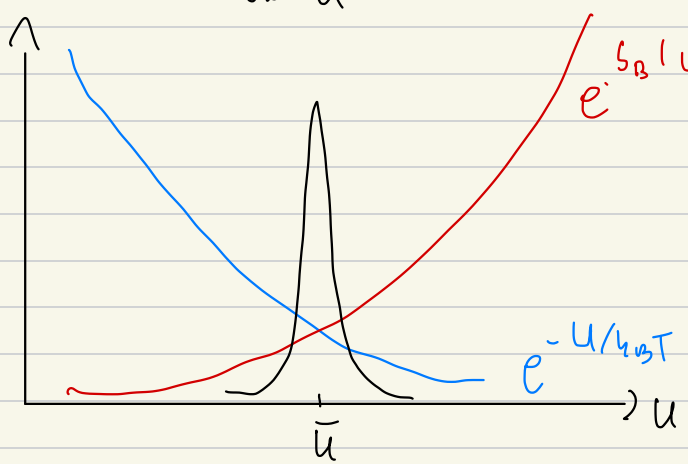
microstates with high energy E_ω are unlikely due to Boltzmann factor e^{-E_ω/k_BT}

but usually # states grows exponentially with E!

$$P(U) = \sum_{\omega:\, E_\omega \approx U} p_\omega = \frac{1}{Z}\,\Omega(U)\, e^{-U/k_BT} = \frac{1}{Z}\, e^{S_B(U)}\, e^{-U/k_BT}$$

$$\downarrow \quad \text{free energy!}$$

$$\frac{1}{Z}\, e^{-(U-TS_B)/k_BT}$$



equilibrium

$\Updownarrow$

max free energy

Entropy of p^{can}: $S_G = -\sum_\omega p_\omega^{can} \ln p_\omega^{can} = -\frac{k_B}{Z}\sum_\omega e^{-\frac{E_\omega}{k_BT}} \ln\left(\frac{e^{-\frac{E_\omega}{k_BT}}}{Z}\right)$

$$= k_B \ln Z + \frac{1}{T}\underbrace{\frac{1}{Z}\sum_\omega e^{-\frac{E_\omega}{k_BT}} E_\omega}_{\bar{U}}$$

$$\Rightarrow \boxed{F = \bar{U} - TS_G \approx -k_BT \ln Z}$$

define $\beta := \frac{1}{k_B T}$

$$\bar{U} = \sum_w E_w p_w^{can} = \sum_w \frac{e^{-\beta E_w}}{Z} E_w = -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = \frac{\partial}{\partial \beta} \ln Z$$

$$\frac{\partial Z}{\partial \beta} = \sum_w e^{-\beta E_w}(-E_w)$$

energy variance (fluctuations):

$$(\delta U)^2 := \overline{U^2} - \bar{U}^2 = \sum_w E_w^2 p_w^{can} - \bar{U}^2 = \ldots = \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$\ln Z = -\beta F \sim N$ (extensive)

$\rightarrow \bar{U} \sim N$, $\delta U \sim \sqrt{N}$ } relative fluctuations: $\frac{\delta U}{\bar{U}} \sim \frac{1}{\sqrt{N}}$

$\rightarrow 0$ as $N \rightarrow \infty$

In the limit $N \rightarrow \infty$, fluctuations become negligible

$\bar{U} \rightarrow U$ thermodynamic internal energy

$\Rightarrow$ canonical and microcanonical ensembles give same results (up to $1/\sqrt{N}$)

1st law:

$$dU = d\left(\sum_w p_w E_w\right) = \underbrace{\sum_w p_w\, dE_w}_{\text{work}} + \underbrace{\sum_w (dp_w) E_w}_{\text{heat}}$$

$$T dS = -k_B T \sum_w d(p_w \ln p_w) = \quad (d\,p\,(\ln p + 1))$$

$$= -k_B T \sum_w dp_w \left(-\frac{E_w}{k_B T} - \ln Z + 1\right)$$

$$= \sum_w E_w\, dp_w$$

$\sum_w p_w = 1 \Rightarrow \sum_w dp_w = 0$

1st term: $dE_w = \frac{\partial E_w}{\partial V} dV + \frac{\partial E_w}{\partial N} N + \dots$

$$dU = \underbrace{\left(\sum_w p_w \frac{\partial E_w}{\partial V}\right)}_{\frac{\partial U}{\partial V}} dV + \left(\sum_w p_w \frac{\partial E_w}{\partial N}\right) dN + T dS$$

3rd law

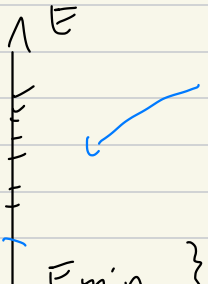
$P_w \sim e^{-E_w/k_B T}$ → as $T \to 0$: only lowest energy states ("ground states")

of g.s. $= \Omega_0$

entropy at $T=0$: $S_0 = k_B \ln \Omega_0$

If g.s. is unique: $\Omega_0 = 1 \to S_0 = 0$ 3rd law

more precisely: we approach $T \to 0$ as a limit



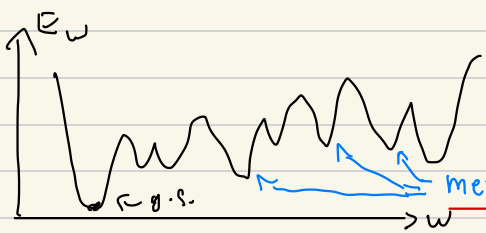
quantum mechanics: discrete energy levels

$\} \leftarrow$ # of states in $[E_{min}, E_{min} + \varepsilon] = \Omega$

In most systems $\Omega_0 \sim N^a$ a small power (0, 1, 2, etc.)

$\to S_0 \sim \ln N$ $\quad S_0/N \to 0$ as $N \to \infty$

Glasses:



equilibration time becomes huge

metastable states

Statistical physics of ideal gas

Ideal gas: no interactions between particles

$$E = \sum_i \frac{p_i^2}{2m} \qquad i = 1, 2, 3, \ldots, N$$

Write $E = N \frac{p_{rms}^2}{2m}$ → defines "root mean square" mom. p_{rms}

↳ "typical" momentum of particles

Boltzmann approach: need Ω

no interactions → x, p cells are independent

$$\Omega_{pos} = \left(V/(\Delta x)^3\right)^N$$

To approximate $\Omega_{mom.}$ ignore $p > p_{rms}$

↳ $\Omega_{mom} \approx (p_{rms}/\Delta p)^{3N}$

Two possibilities:

- distinguishable particles: $\Omega = \Omega_{pos} \cdot \Omega_{mom}$
- indistinguishable: $\Omega = \Omega_{pos} \cdot \Omega_{mom} / N!$

↳ we will use this (in quantum mechanics, indistinguishability is fundamental)

Stirling approximation:

$$\ln(N!) = \ln(1) + \ln(2) + \dots + \ln(N)$$

$$= \sum_{j=1}^{N} \ln(j) \approx \int_1^N \ln x \, dx = \left[x \ln x - x \right]_1^N$$

$$= N \ln N - N \Rightarrow \boxed{N! \approx (N/e)^N}$$

$$\Omega \approx \left(\frac{eV p_{rms}^3}{N (\Delta x \Delta p)^3} \right)^N = \frac{eV (2mU)^{3/2}}{N^{5/2} h^3}$$

$\Delta x \Delta p \approx h$ (Planck's const.)

$N = nN_A$

$$S_B = k_B \ln \Omega \approx n k_B N_A \left(\ln \frac{V}{n} + \frac{3}{2} \ln \frac{U}{n} + \text{const.} \right)$$

v

ideal gas: $\frac{U}{n} \sim T$

$$\rightarrow s = \frac{S}{n} = k_B N_A \left(\ln v + \frac{3}{2} \ln T \right) + s_0$$

previously we had $s = c_V \ln T + R \ln V + s_0$

c_V: heat capacity / mol

consistency requires $R = k_B N_A$; $c_V = 3R/2$

pressure: $P = T \left(\frac{\partial S}{\partial V} \right)_{U,N} = T n k_B N_A / V \approx \frac{nRT}{V}$ ✔

Gibbs approach:

size of cells $= h$

microstate $\omega: \{\underline{x}_i, \underline{p}_i\} \to \sum_\omega = \frac{1}{h^{3N}} \int d^3x_1 \ldots \int d^3p_N$

$$Z = \frac{1}{h^{3N}} \int d^3x_1 \ldots d^3x_N \int d^3p_1 \ldots d^3p_N \; e^{-\frac{1}{k_BT}\sum_i \frac{p_i^2}{2m}}$$

$$= V^N \left(\underbrace{\frac{1}{h^3} \int d^3p \, e^{-\frac{1}{k_BT}\frac{p^2}{2m}}}_{Z_{1p}} \right)^N$$

Z_{1p} part fc. for 1 particle

For indistinguishable particles:

$$Z = \frac{1}{N!} V^N Z_{1p}^N \approx \left(eV Z_{1p}/N\right)^N$$

$$Z_{1p} = \frac{4\pi}{h^3} \int_0^\infty dp \, p^2 \, e^{-\frac{p^2}{2mk_BT}} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}$$

Free energy: $F = -k_BT \ln Z = -Nk_BT \ln\left[\frac{eV}{N}\left(\frac{2\pi m k_BT}{h^2}\right)^{\frac{3}{2}}\right]$

$\hookrightarrow p = -\frac{\partial F}{\partial V} = \frac{Nk_BT}{V}$ ✓

$$S_G = -\frac{\partial F}{\partial T} = Nk_B \ln\left[\frac{eV}{N}\left(\frac{2\pi m k_BT}{h^2}\right)^{\frac{3}{2}}\right] + Nk_BT \, \frac{3}{2} \frac{1}{T}$$

$$= Nk_B \ln\left(\frac{e^{5/2}\left(\frac{4\pi}{3} m u\right)^{3/2}}{N^{5/2} h^2}\right)$$

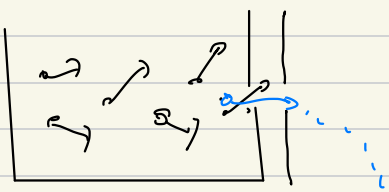
$u = \frac{3}{2} k_BT$

Kinetic theory

Microscopic perspective can go beyond thermo and give new insights

E.g.: What is the probability of a given particle having speed v?

How to measure:



$\underline{v}_j$: velocity of j-th particle

↳ $g(\underline{v}_j)$ some function e.g. $v_{j,x}$ v_j^2, ...

$$\overline{g(\underline{v}_j)} = \frac{1}{\mathcal{Z}} \int d^3r_1 \dots \int d^3r_N \int d^3v_1 \dots d^3v_N \; e^{-\sum_n \frac{m v_n^2}{2k_BT}} g(\underline{v}_j)$$

$$= \frac{\int d^3v_j \, g(\underline{v}_j) \, e^{-\frac{m v_j^2}{2k_BT}}}{\int d^3v_j \, e^{-\frac{m v_j^2}{2k_BT}}} \equiv \int d^3v_j \, p_{1p}(\underline{v}_j) \, g(\underline{v}_j)$$

$$p_{1p}(v_x, v_y, v_z) = \frac{1}{\mathcal{Z}_{1p}} e^{-\frac{m}{2k_BT}(v_x^2 + v_y^2 + v_z^2)}$$

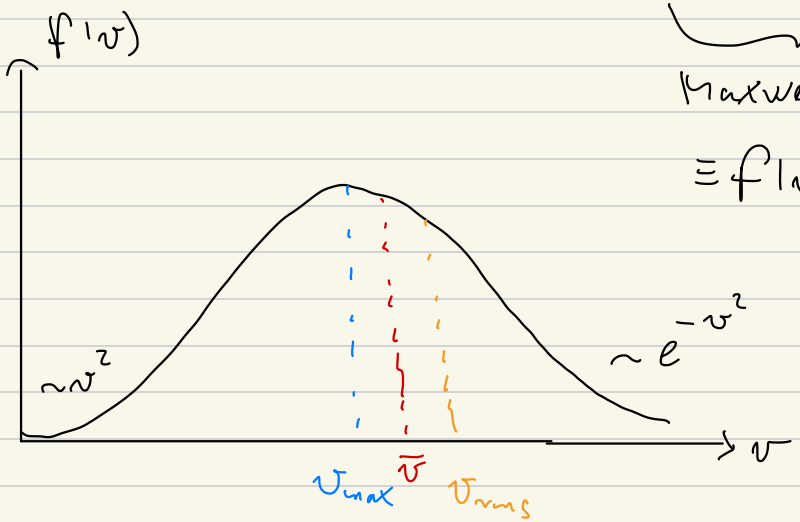
↳ $\overline{v_x} = \overline{v_y} = \overline{v_z}$ $(\underline{v}_j \to \underline{v})$ (spherically symmetric)

what if $g(\underline{v}) = g(v)$?

$$\int d^3v\, g(v)\, e^{-\frac{mv^2}{2k_BT}} = 4\pi \int dv\, v^2\, e^{-\frac{mv^2}{2k_BT}}\, g(v)$$

$$\rightarrow p_{1p}(v) = \frac{4\pi v^2 e^{-\frac{mv^2}{2k_BT}}}{Z_{1p}} = \underbrace{\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} 4\pi v^2 e^{-\frac{mv^2}{2k_BT}}}_{\text{Maxwell (-Boltzmann) dist.}}$$

$\equiv f(v)$

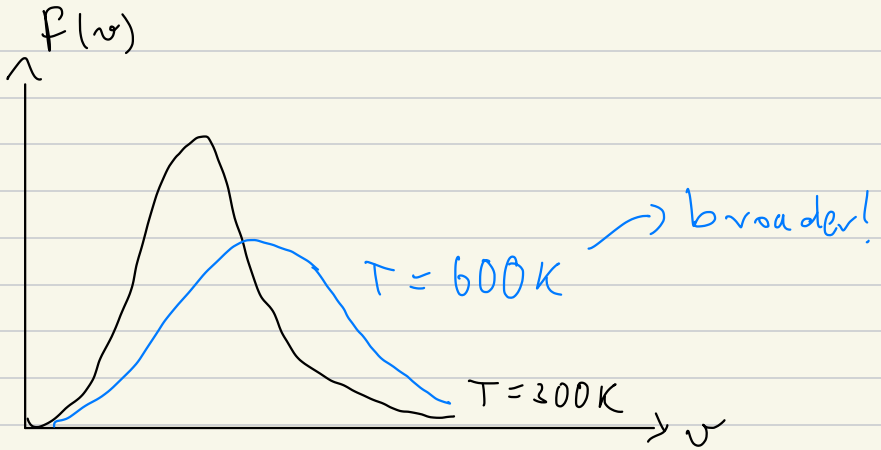


$$v_{max} = \sqrt{\frac{2k_BT}{m}} \qquad \left(\left.\frac{df}{dv}\right|_{v_{max}} = 0\right)$$

$$\bar{v} = \int dv\, f(v)\, v = \sqrt{\frac{8k_BT}{\pi m}} = \frac{2}{\sqrt{\pi}} v_{max}$$

$$v_{rms} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3}{2}}\, v_{max}$$

} ratios are independent of T, m



At room temperature for nitrogen molecule: $\bar{v} \approx 500 \frac{m}{s}$

helium: $\bar{v} \approx 1300 \frac{m}{s}$

note: escape velocity: $v_{esc} = \sqrt{\frac{2GM}{R}}$

if M too small / R too big → no atmosphere!

For Earth: $v_{esc} \approx 11000 \frac{m}{s} \gg \bar{v}_{He}$

but in fact there is still He escaping from tail of f(v)

Barometric formula

So far, we ignored gravitational field

In fact: $E = \sum_j \frac{m v_j^2}{2} + \sum_j m g z_j$

↳ probability distribution for z coordinate:

$$p(z) = \frac{e^{-mgz/k_B T}}{\int dz\, e^{-mgz/k_B T}}$$

↓

density of gas decreases as $\rho(z) \sim e^{-mgz/k_B T}$

$\rho = N/V$

Collisions, diffusion, viscosity

$\bar{v} \approx 500 \frac{m}{s}$ too fast. Why?

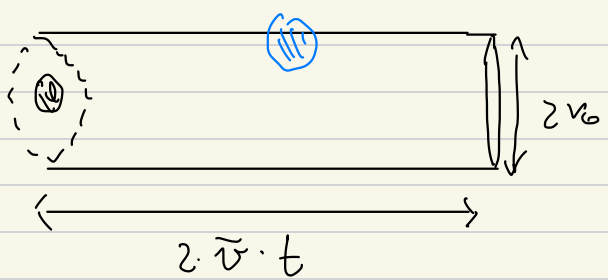
↳ collisions!



How often does this happen?

model molecule as ball of radius r_0

collision happens if there is another molecule within distance $2r_0$



density $= N/V$

volume: $8\pi r_0^2 \bar{v} t \rightarrow N_{coll} \approx 8\pi r_0^2 \bar{v} t \cdot \rho$

$\bar{v} \approx 10^5 \frac{cm}{s}$

$\rho \approx 10^{19} \frac{1}{cm^3}$

$r_0 \approx 10^{-8} cm$

$\rightarrow \frac{N_{coll}}{t} \approx 10^9 \frac{1}{s}$ Very frequent!

Average time between collisions: $\tau = 10^{-9}\,s$

↳ <u>scattering time</u>

How far a particle travels between two collisions:

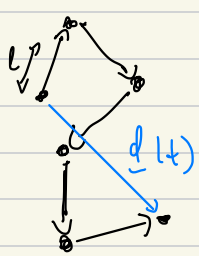
$$\ell = \bar{v}\tau = \frac{1}{8\pi r_0^2 \rho} \simeq 10^{-4} - 10^{-5}\ cm$$

note: $\ell \approx 10^3 r_0 >> r_0$ → dilute gas

How to describe particle's motion on scales $t >> \tau$?

Simple model: each collision completely randomizes the direction

Take time/distance between collisions to be exactly τ, ℓ



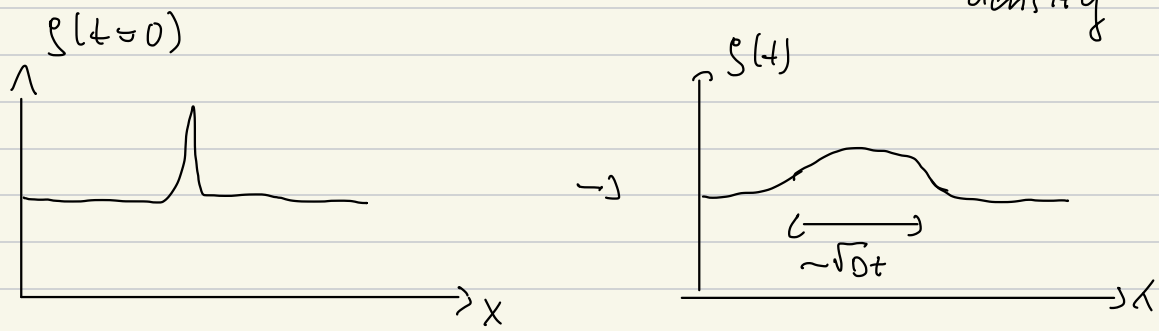
$$\underline{d}(t) \simeq \sum_{i=1}^{t/\tau} \underline{d}_i \quad \text{where } |\underline{d}_i| = \ell$$

$$\overline{\underline{d}_i} = 0 \ \text{(random direction)} \rightarrow \overline{\underline{d}} = 0$$

$$\overline{d^2} = \sum_{i,j} \overline{d_i \cdot d_j} = \sum_i \overline{d_i^2} + \sum_{i \neq j} \overline{d_i} \cdot \overline{d_j} = \frac{t}{\tau}\ell^2 + 0$$

↳ $d_{rms} := \sqrt{\overline{d^2}} = \sqrt{\frac{\ell^2}{\tau} t}$ <u>diffusive</u>; diffusion const: $D = \frac{\ell^2}{2\tau}$

More macroscopic picture: Inhomogeneities in the density



example: ink spreading in water

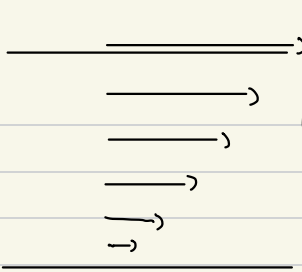
diffusion equation:

$$\rho(t+\Delta t, x) = \rho(t, x-\Delta x) = \rho(t,x) - \frac{\partial \rho}{\partial x}\overline{\Delta x} + \frac{1}{2}\frac{\partial^2 \rho}{\partial x^2}\overline{\Delta x^2}$$

($\overline{\Delta x} = 0$)

$$\downarrow$$

$$\partial_t \rho = \frac{1}{2}\frac{\overline{\Delta x^2}}{\Delta t}\frac{\partial^2 \rho}{\partial x^2} = D\frac{\partial^2 \rho}{\partial x^2}$$



u(z) speed distribution

Viscosity:

Friction between layers: $F = A \eta \frac{du}{dz}$

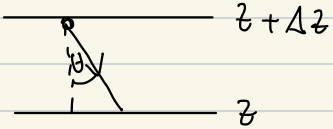
↳ viscosity of fluid

consider a layer at height z

Force = change in momentum due to collisions

when colliding with a particle coming from $z + \Delta z$

$$\Delta p = m\left(u(z+\Delta z) - u(z)\right) \approx m \frac{du}{dz} \cdot \Delta z = m \frac{du}{dz} \ell \cos\theta$$



How many collisions per unit time?

↳ need to count particles within distance $v_z \cdot \Delta t$ in the z direction

$$\rightarrow \rho \cdot A \cdot v_z \cdot \Delta t = N_{coll}$$

$$F = \frac{\Delta P_{tot}}{\Delta t} = \frac{N_{coll} \cdot \Delta p}{\Delta t} = A \rho \int d^3v \, P_{lp}(\underline{v}) \, v_z \, \Delta p$$

polar coordinates: $\underline{v} = v \begin{pmatrix} \sin\theta\cos\ell \\ \sin\theta\sin\ell \\ \cos\theta \end{pmatrix}$
(spherical)

$$d^3v = v^2 \sin\theta \, dv \, d\theta \, d\ell$$

$$F = A m \varrho \frac{du}{dz} \ell \underbrace{\int_0^{2\pi} d\ell}_{2\pi} \underbrace{\int_0^{\pi} d\theta \, \sin\theta \cos^2\theta}_{2/3} \underbrace{\int_0^{\infty} dv \, v \cdot \frac{f(v)}{4\pi}}_{\bar{v}/4\pi}$$

$$= A \frac{du}{dz} \frac{m \varrho \ell \bar{v}}{3}$$

$$\Rightarrow \boxed{\eta = \frac{1}{3} m \varrho \ell \bar{v}}$$

but: $\ell \sim 1/\varrho \rightarrow \eta$ is independent of ℓ!

Van der Waals equation of state:

ideal gas: $pV = Nk_BT$ ignores interactions

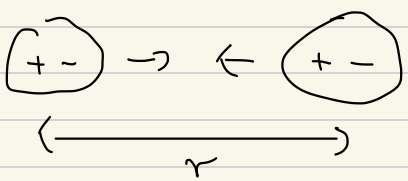
but we saw that particles have finite radius

when we calculate Z: $\int d^3x_i = V - N\frac{4\pi}{3}r^3$

↳ replace V with $V - bN$ "effective volume"

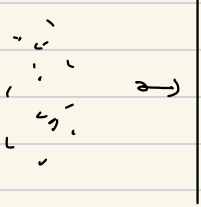
In fact, there is also a long-range attraction

$$E = \sum_j \frac{mv_j^2}{2} + \frac{1}{2}\sum_{i\neq j} u(|\underline{x}_i - \underline{x}_j|)$$



weak attractive force (goes as $1/r^6$)

This decreases the pressure (pulls particle away from wall)



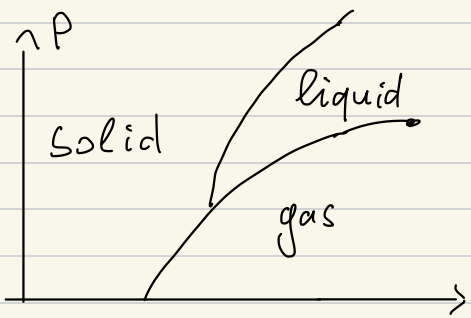
Since all pairs of particles interact, this effect $\sim N^2$

$$\Rightarrow P \simeq \frac{Nk_BT}{V - bN} - a\left(\frac{N}{V}\right)^2 \simeq \frac{\rho}{1-b\rho}k_BT - a\rho^2$$

Phase transitions

As we change T, P, system can suddenly rearrange into qualitatively different structures

e.g. water:



Many other phases!

E.g. plasma, superfluid, superconductor, ferromagnet

also: different crystal structures (e.g. ice)

↳ wikipedia / Phases_of_ice

Transitions between phases are <u>sharp</u>

↳ physical quantities jump / diverge

(e.g. entropy, specific heat etc.)

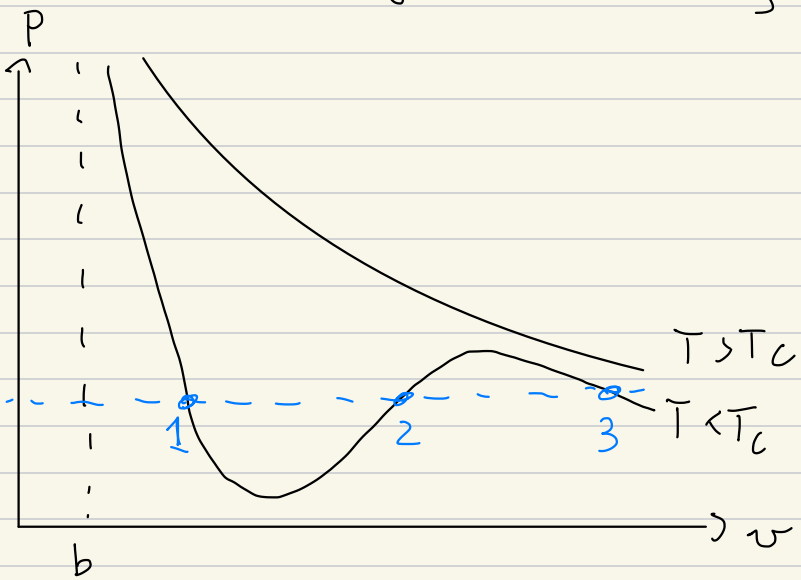
(note: lot of interesting current research on understanding new phases and phase transitions especially at low T where quantum effects become important)

↳ e.g. superfluid, superconductor

Case study: liquid - gas transition

van der Waals: $P = \frac{k_B T}{v - b} - \frac{a}{v^2}$ $\quad v = \frac{V}{N} = \frac{1}{\rho} > b$

Isotherms:



At low T: multiple solutions with same T, P

1: v is small ($\approx b$) → high density

compressibility: $\frac{1}{K} = \frac{-1}{V}\left(\frac{dP}{dV}\right)_T$ is big → incompressible

⇒ liquid state

3: v is big → low density

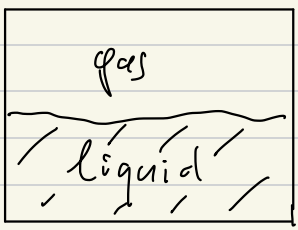
K is big → compressible

} ⇒ gas state

What about state 2?

$\frac{dP}{dV} > 0$: negative compressibility → unstable!

Instead we find phase separation:



When are the liquid and gas in equilibrium with each other?

↳ need: $P_{gas} = P_{liquid}$, $T_{gas} = T_{liquid}$ ✓

$\mu_{gas} = \mu_{liquid}$

$\mu = \frac{G}{N}$ Gibbs free energy / particle

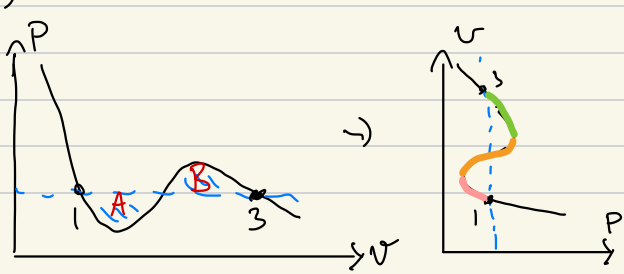
Moving along isotherm: $d\mu = \left(\frac{\partial \mu}{\partial P}\right)_T dP = v\, dP$

$\frac{1}{N}\left(\frac{\partial G}{\partial P}\right)_T = \frac{V}{N} = v$

$\mu_{gas} - \mu_{liquid} = \int_1^3 dP' v(T, P')$

In equilibrium: areas above/below must be equal

$A = B$

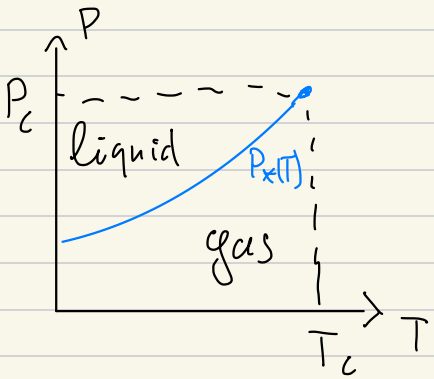


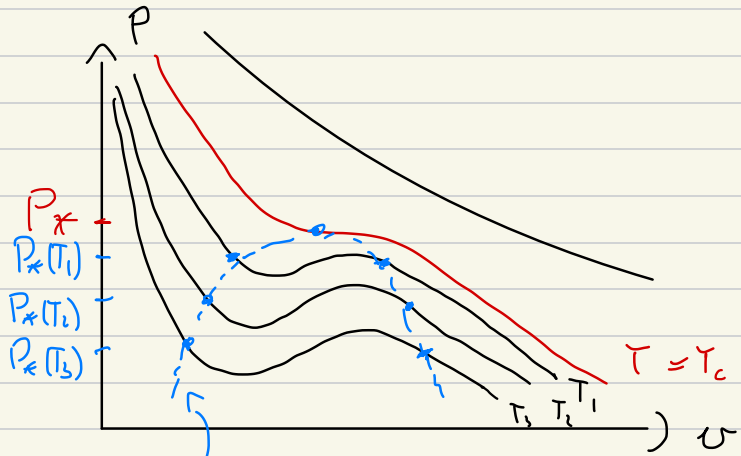
$\mu_{gas} = \mu_{liquid} \rightarrow$ fixes pressure $P = P_*(T)$

$P < P_*(T)$: $\mu_{gas} < \mu_{liquid}$

$(G_{gas} < G_{liquid}) \rightarrow$ gas is favored

$P > P_*(T)$: $\mu_{liquid} < \mu_{gas} \rightarrow$ liquid is favored

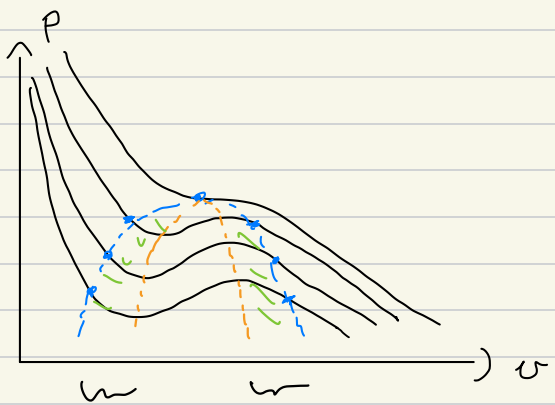




At $P = P_*(T)$: we can freely distribute particles between liquid and gas states

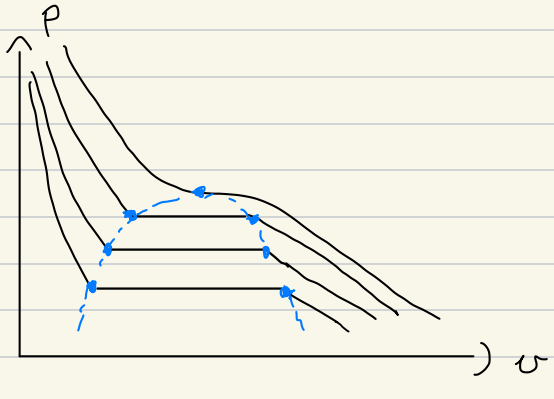
$$v = \frac{V}{N} = \frac{N_{liq}}{N_{liq} + N_{gas}} v_{liq} + \frac{N_{gas}}{N_{liq} + N_{gas}} v_{gas}$$

→ v can be anywhere between v_{liq} and v_{gas}



metastable liquid

metastable gas

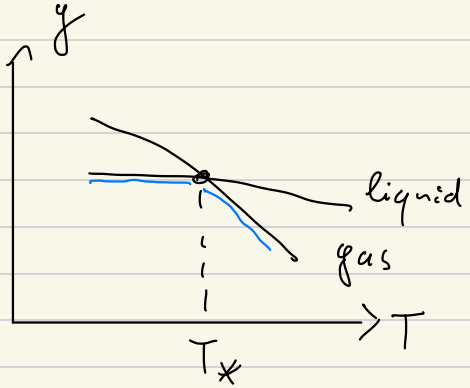


"Maxwell construction"

have $\kappa > 0$ but still ultimately non stable

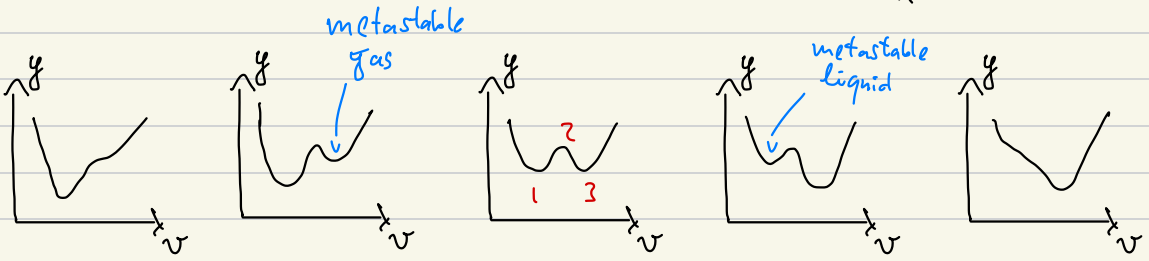
Similar phenomenon in freezing: supercooled liquid

Let $g := \frac{G}{N}$



At $T > T_*$: $g_{gas} < g_{liq}$

At $T < T_*$: $g_{gas} > g_{liq}$



$T \ll T_*$ $T < T_*$ $T = T_*$ $T > T_*$ $T \gg T_*$

At T_*: slope $\frac{\partial g}{\partial T}$ changes discontinuously

$\rightsquigarrow s$ = entropy per particle: $\Delta s = s_{gas} - s_{liq} \neq 0$

Related to <u>latent heat</u>: $L = T\Delta S = N \underbrace{T \Delta s}_{\Delta \ell}$

Similarly in g as a function of $P \rightarrow v_{gas} - v_{liq} \neq 0$

What is the shape of $P_*(T)$?

↳ defined by $g_{gas} = g_{liq}$ $\quad g = G/N$

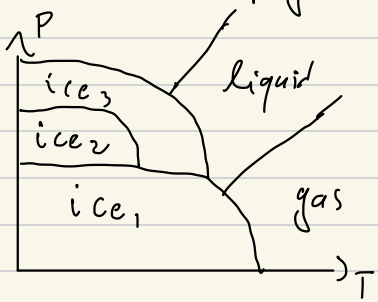
along this line: $dg_{gas} = dg_{liq}$

$$dg = -s\,dT + v\,dP$$

$$\Rightarrow (v_{gas} - v_{liq})\,dP \overset{!}{=} (s_{gas} - s_{liq})\,dT$$

$$\rightarrow \frac{dP}{dT} = \frac{\Delta s}{\Delta v} = \frac{\ell}{T\cdot\Delta v} = \frac{L}{T\cdot\Delta V}$$

Clausius - Clapeyron



3rd law: $\Delta S \to 0 \Rightarrow \frac{dP}{dT} \to 0$ as $T \to 0$

Approximate solution:

assume L is independent of T, P

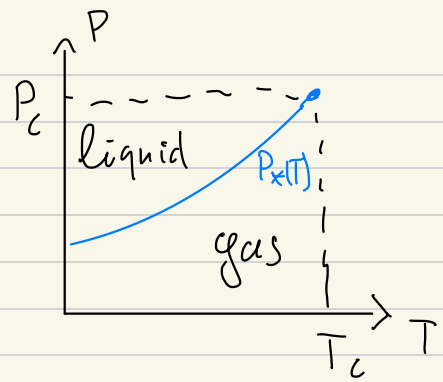
assume $v_{gas} \gg v_{liq}$ (for water: $v_{gas}/v_{liq} \approx 10^3$)

v_{gas} is big → approximate as ideal gas: $P v_{gas} = R\cdot T$

$$\rightarrow \frac{dP}{dT} \approx \frac{\ell}{T v_{gas}} = \frac{\ell P}{RT^2} \rightarrow \frac{dP}{P} = d(\ln P) = \frac{\ell}{R}\frac{dT}{T^2}$$

$$\rightarrow \ln P = -\frac{\ell}{RT} + \text{const.} \rightarrow \underline{P = \text{const} \times \exp(-\ell/RT)}$$

Critical point (P_c, T_c)



As we approach it:

$\Delta s, \Delta v \to 0$

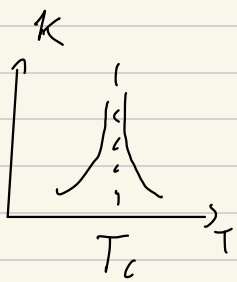
$\to$ no jump in $\frac{\partial g}{\partial T}, \frac{\partial g}{\partial P}$

but: divergences in 2nd derivatives

e.g. specific heat: $c = \frac{\partial^2 g}{\partial T^2} \sim \frac{\partial s}{\partial T}$

compressibility $\kappa = \frac{\partial^2 g}{\partial P^2} \sim \frac{\partial v}{\partial P}$

e.g. $\kappa \approx |T - T_c|^{-\gamma}$ when $T \approx T_c$



γ: <u>critical exponent</u>

van der Waals equation predicts $\gamma = 1$

In fact: $\gamma \approx 1.2$

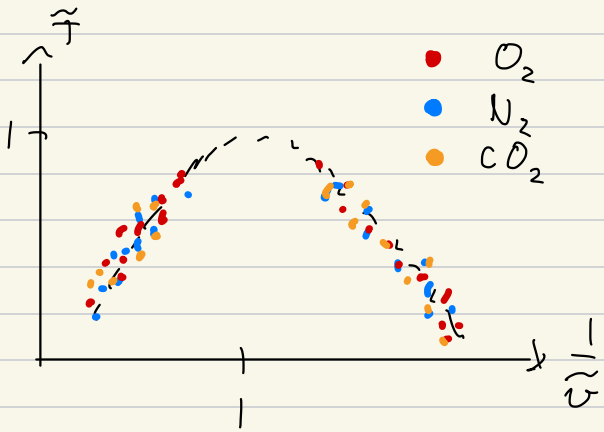
$\Downarrow$

2nd order phase transition

Universality: close to the critical point, properties become independent of what kind of gas we consider

reduced variables: $\tilde{T} = \frac{T}{T_c}$, $\tilde{p} = \frac{p}{p_c}$, $\tilde{v} = \frac{v}{v_c}$

coexistence curve:

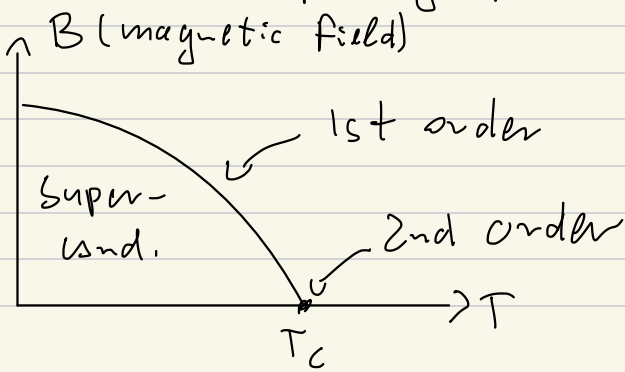


Same critical exponents appear in completely different systems, e.g. paramagnet ~ ferromagnet phase transition

— o —

Another example of phase transitions: superconductor

has 0 resistance



Quantum Statistics

Reminder: for ideal gas we found $C_V = \text{const.}$

↳ contradicts 3rd law: $\lim_{T \to 0} C_V = 0$

As $T \to 0$: need to use quantum mechanics

quantum: energy levels are quantized

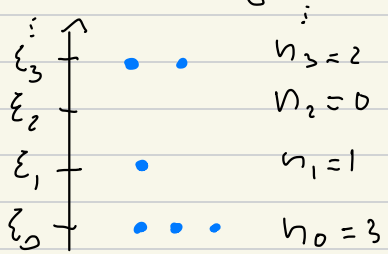
ideal gas: particle energies are independent

energy levels of one particle: ε_j

microstates: How many particles occupy level ε_j → n_j

$$E(\underline{n}) = \sum_j \varepsilon_j \cdot n_j$$

$$\underline{n} = (n_0, n_1, n_2, \ldots)$$



2 types of particles: bosons and fermions

fermions: cannot occupy same state (Pauli exclusion)

↳ $n_j = 0, 1$

for bosons: $n_j = 0, 1, 2, 3, \ldots$

Fermions:

number of particles: N

$f_j(N) =$ probability that $n_j = 1$

note: $\bar{n}_j = 1 \cdot f_j + 0 \cdot (1 - f_j) = f_j$

Divide microstates into two groups:

$A(N)$: configurations where $n_j = 1$

$B(N)$: " " " $n_j = 0$

partition fc: $Z_N = \sum_{\underline{n}} e^{-\beta E(\underline{n})}$ $\qquad \beta = \frac{1}{k_B T}$

$$f_j(N) = \sum_{\underline{n} \in A(N)} \frac{e^{-\beta E(\underline{n})}}{Z_N} = 1 - \sum_{\underline{n} \in B(N)} \frac{e^{-\beta E(\underline{n})}}{Z_N}$$

trick: if we take any $\underline{n} \in B(N)$ and add a particle with energy ε_j, we get a state in $A(N+1)$

$$\sum_{\underline{n} \in B(N)} \frac{e^{-\beta E(\underline{n})}}{Z_N} \simeq \sum_{\underline{n}' \in A(N+1)} \frac{e^{-\beta (E(\underline{n}') - \varepsilon_j)}}{Z_N} = e^{\beta \varepsilon_j} \frac{Z_{N+1}}{Z_N} \underbrace{\sum_{\underline{n}' \in A(N+1)} \frac{e^{-\beta E(\underline{n}')}}{Z_{N+1}}}_{f_j(N+1)}$$

$$Z_N = e^{-\beta F_N} \rightarrow \frac{Z_{N+1}}{Z_N} = e^{-\beta(F_{N+1} - F_N)}$$

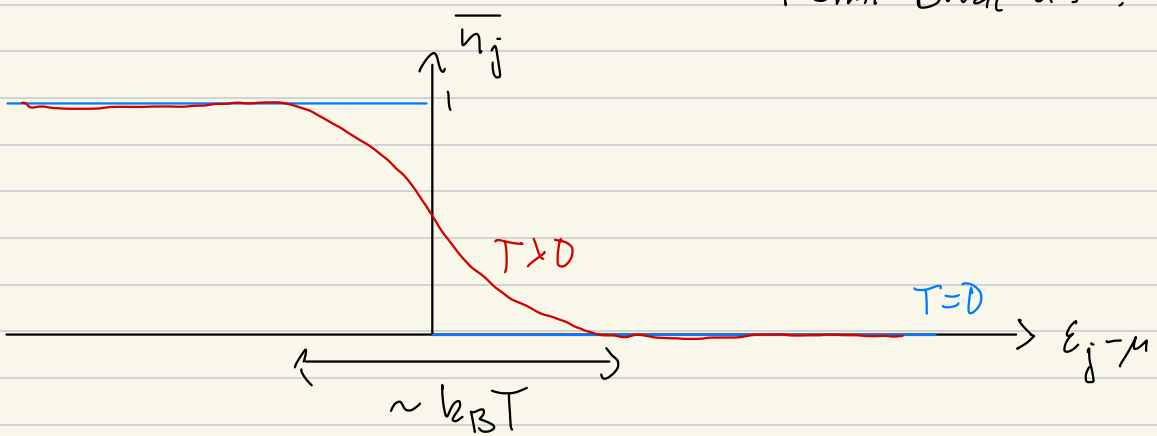
$$F_{N+1} - F_N \approx \left(\frac{\partial F}{\partial N}\right)_{T,V} = \mu \quad \text{chemical potential}$$

$$\Rightarrow f_j(N) = 1 - e^{\beta(\varepsilon_j - \mu)} f_j(N+1)$$

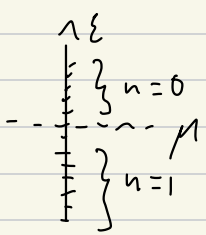
$$N \gg 1 \rightarrow f_j(N) \approx f_j(N+1)$$

$$\Rightarrow \boxed{\overline{n_j} = \frac{1}{e^{\beta(\varepsilon_j - \mu)} + 1}}$$

Fermi-Dirac distr.



$T = 0$: step fct:



$\lim_{T \to 0} \mu = \varepsilon_F$ Fermi energy

$$U(T) - U(0) \sim (k_B T) \cdot (k_B T) \rightarrow C_V \sim \left(\frac{\partial U}{\partial T}\right)_{V,N} \sim T$$

(excited particles) · (typical energy)

valid for metals at low T

why did μ show up?

If system can exchange particles with environment

$$e^{-\beta E} \to e^{-\beta(E-\mu N)}$$ "grand canonical" distr.

for a single particle: $p(n_j) = \frac{1}{z_{1p}} e^{-\beta(\varepsilon_j n_j - \mu n_j)}$

$$\downarrow \qquad z_{1p} = \sum_{n_j} e^{-\beta(\varepsilon_j - \mu) n_j}$$

$$\overline{n_j} = \frac{\sum_{n_j} n_j e^{-\beta(\varepsilon_j - \mu) n_j}}{\sum_{n_j} e^{-\beta(\varepsilon_j - \mu) n_j}}$$

Fermions: $n_j = 0, 1 \to \overline{n_j} = \frac{e^{-\beta(\varepsilon_j - \mu)}}{1 + e^{-\beta(\varepsilon_j - \mu)}} = \frac{1}{e^{\beta(\varepsilon_j - \mu)} + 1}$

Bosons:

$$\sum_{n_j} n_j \, e^{-\beta(\varepsilon_j-\mu)n_j} = \frac{1}{\beta}\frac{\partial}{\partial\mu}\underbrace{\sum_{n_j} e^{-\beta(\varepsilon_j-\mu)n_j}}_{z_{1p}}$$

$$z_{1p} = \sum_{n_j=0}^{\infty} e^{-\beta(\varepsilon_j-\mu)n_j} = \frac{1}{1-e^{-\beta(\varepsilon_j-\mu)}}$$

geometric series: $1+x+x^2+\dots$ needs $x<1 \to \mu < \varepsilon_j$

$$\overline{n_j} = \frac{1}{\beta}\frac{\partial}{\partial\mu}\ln z_{1p} = -\frac{1}{\beta}\frac{\partial}{\partial\mu}\ln\left(1-e^{-\beta(\varepsilon_j-\mu)}\right)$$

$$= -\frac{1}{\beta}\,\frac{-e^{-\beta(\varepsilon_j-\mu)}\,\beta}{1-e^{-\beta(\varepsilon_j-\mu)}} = \boxed{\frac{1}{e^{\beta(\varepsilon_j-\mu)}-1}}$$

Bose-Einstein distribution

note: μ itself depends on T; $\mu \to \varepsilon_0 = 0$ as $T \to 0$

$\overline{n_j}$ diverges as $\mu \to \varepsilon_0$ ↝ signals a phase transition to a Bose-Einstein condensate

$\left(e^{\beta\mu} \approx 1 - \frac{\text{const.}}{N}\right)$ $\overline{n_0} \sim N$ ↓ macroscopic occupation

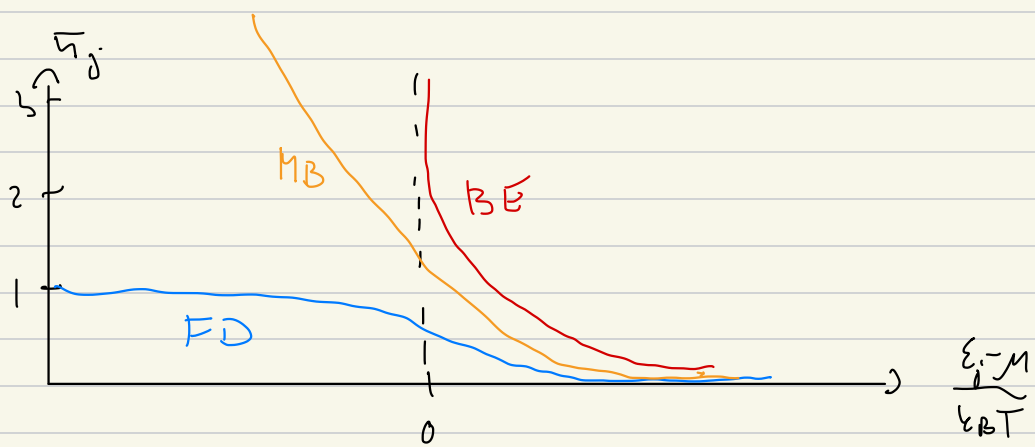
$$\bar{n}_j = \frac{1}{e^{\beta(\varepsilon_j - \mu)} \pm 1}$$

Classical limit: $\bar{n}_j \ll 1$

$\rightarrow e^{\beta(\varepsilon_j - \mu)} \pm 1 \gg 1$ to apply for all ε_j: want $e^{\beta\mu} \ll 1$

$\rightarrow e^{\beta(\varepsilon_j - \mu)} \gg 1$

$\rightarrow \bar{n}_j \approx e^{-\beta(\varepsilon_j - \mu)}$ Maxwell - Boltzmann



for fermions: classical when $k_B T \gg \varepsilon_F$

but in typical metals: $\varepsilon_F / k_B \approx 10^4 - 10^5$ K !

↳ still very much quantum at room temperature